

MdZAT5 regulates drought tolerance via mediating accumulation of drought-responsive miRNAs and mRNAs in apple

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Summary

- Drought limits apple yield and fruit quality. However, the molecular mechanism of apple in response to drought is not well known. Here, we report a Cys2/His2 (C2H2)-type zinc-finger protein, MdZAT5, that positively regulates apple drought tolerance by regulating drought-responsive RNAs and microRNAs (miRNAs).
- DNA affinity purification and sequencing and yeast-one hybrid analysis identified the binding motifs of MdZAT5, T/ACACT/AC/A/G. Chromatin immunoprecipitation quantitative polymerase chain reaction (ChIP-qPCR) and electrophoretic mobility shift assays (EMSAs) showed that MdZAT5 directly binds to the promoters of the drought-responsive genes including *MdRHA2a*, *MdLEA14*, *MdTPX1*, and *MdCAT3*, and activates their expression under drought stress.
- MdZAT5 interacts with and directly targets HYPONASTIC LEAVES1 (MdHYL1). MdZAT5 may facilitate the interaction of MdHYL1 with pri-miRNAs or MdDCL1 by activating *MdHYL1* expression, thereby regulating the biogenesis of drought-responsive miRNAs. Genetic dissection showed that *MdHYL1* is essential for MdZAT5-mediated drought tolerance and miRNA biogenesis. In addition, ChIP-qPCR and EMSA revealed that MdZAT5 binds directly to the promoters of some MIR genes including *Mdm-miR171i* and *Mdm-miR172c*, and modulates their transcription.
- Taken together, our findings improve our understanding of the molecular mechanisms of drought response in apple and provide a candidate gene for the breeding of drought-tolerant cultivars.

Introduction

Drought is one of the most serious threats limiting plant growth and crop yield (Zhu, 2016). In recent years, climate change has led to frequent drought events, resulting in reduced yields and quality of crops (Gupta *et al.*, 2020). It is, therefore, necessary to breed new drought-tolerant varieties to alleviate the adverse effects of drought and increase crop yield and quality. Plants are sessile organisms and have evolved many defense mechanisms to adapt to drought conditions. In particular, reactive oxygen species (ROS) metabolism plays a key role in the acclimation of plants to drought stress (Choudhury *et al.*, 2017; Laxa *et al.*, 2019). Under drought stress, the overproduction of ROS disrupts the electron transport chain and impairs multiple cellular processes, thereby causing serious damage to plants (Gill & Tuteja, 2010). However, this process can be alleviated by enzymatic and nonenzymatic antioxidants, including peroxidase (POD), superoxide dismutase (SOD), catalase (CAT), ascorbate peroxidase (APX), ascorbic acid, and glutathione (GSH) (Gill & Tuteja, 2010; Choudhury *et al.*, 2017; Laxa *et al.*, 2019). The *Arabidopsis cat3* mutant that has lower catalase activity

shows a higher accumulation of hydrogen peroxide (H₂O₂) compared with wild-type (WT) plants and is hypersensitive to drought stress; by contrast, the overexpression of *CAT3* increases the survival rate after rewatering (Zou *et al.*, 2015).

As crucial regulators of post-transcriptional gene regulation, 20–24 nucleotide microRNAs (miRNAs) function through the translational repression or degradation of their mRNA targets (Jones-Rhoades *et al.*, 2006; Li & Zhang, 2016; Song *et al.*, 2019). The miRNAs play important roles in plant development and abiotic stress responses (Song *et al.*, 2019). In *Arabidopsis*, plants that overexpress miR169 are more sensitive to drought stress than WT plants, as their target gene *AtNFYA5* plays a positive role in drought tolerance (Li *et al.*, 2008). In alfalfa, miR156-overexpressing plants have higher stomatal conductance than the WT during drought stress (Arshad *et al.*, 2017). Overexpression of *mdm-miR156* increases the drought tolerance of apple calli (Niu *et al.*, 2019), and overexpression of miR408 increases the drought tolerance of chickpea (Hajyzadeh *et al.*, 2015). Recently, heterologous expression of the rice (*Oryza sativa* L.) miR408 gene in perennial ryegrass conferred drought tolerance by maintaining a higher

leaf relative water content, lower electrolyte leakage, and higher antioxidant capacity compared with the WT (Hang *et al.*, 2021); overexpression of *mdm-miR160e* facilitates apple drought tolerance by targeting *MdARF17* (Shen *et al.*, 2022). Nevertheless, little is known about the regulatory factors of miRNAs under drought stress, particularly in apple.

Early studies identified key proteins of plant miRNA biogenesis, including RNA polymerase II (Pol II), DICER-LIKE1 (DCL1), HUA ENHANCER1 (HEN1), and ARGONAUTE1 (AGO1) (Dong *et al.*, 2008; Zheng *et al.*, 2009; Yu *et al.*, 2010; Arribas-Hernández *et al.*, 2016). During the processing of miRNA precursors, DCL1 forms a dicing complex with the double-stranded RNA (dsRNA) binding domain protein HYPONASTIC LEAVES1 (HYL1/DRB1) and the zinc finger protein SERRATE (SE) (Dong *et al.*, 2008). As a miRNA processing factor, HYL1 is a key regulator of plant development and abiotic stress responses (Vazquez *et al.*, 2004; Liu *et al.*, 2011; Li *et al.*, 2012; Lian *et al.*, 2013). In apple, *MdHYL1* plays a crucial role in drought tolerance and miRNA biogenesis (Shen *et al.*, 2022).

In addition to the ROS scavenging system and miRNAs, transcription factors also play crucial roles in abiotic stress responses. Unlike functional proteins, transcription factors always act in an upstream position in the signal transduction and gene regulatory network, regulating a wide spectrum of downstream genes by binding to specific *cis*-elements in their promoters (Singh *et al.*, 2002). Cys2/His2 (C2H2)-type zinc-finger proteins (ZFPs) form one of the largest transcription factor families in plants and contain a unique conserved QALGGH motif. A considerable number of ZFPs have been reported to regulate plant adaptation to abiotic stress (Englbrecht *et al.*, 2004; Y. Chen *et al.*, 2020). For example, *TaZFP1B* and *OsTZF5* play positive roles in drought tolerance in plants while *DST* negatively regulates drought tolerance (X. Y. Huang *et al.*, 2009; Cheuk *et al.*, 2020; Selvaraj *et al.*, 2020). Among these three ZFPs, *DST* and *TaZFP1B* are two C2H2 ZFPs, while *OsTZF5* is a CCCH-tandem ZFP. Previous RNA-sequencing (RNA-Seq) data revealed that *MdZAT5*, a C2H2 ZFP gene, was induced by simulated drought (Geng *et al.*, 2019). *MdZAT5* positively regulates anthocyanin accumulation and plays a negative regulatory role in salt resistance (Wang *et al.*, 2022). However, whether *MdZAT5* regulates drought tolerance and miRNA biogenesis remains elusive.

In this study, we report that *MdZAT5*, a homolog of *Arabidopsis* *ZAT5*, functions positively in the apple drought response. We showed that *MdZAT5* positively regulates apple drought tolerance by regulating the expression of drought-responsive genes and miRNAs under drought conditions. *MdZAT5* interacts with *MdHYL1* and may contribute to the interaction of *MdDCL1*-*MdHYL1* complex and the interaction of pri-miRNAs with *MdHYL1* under drought stress.

Materials and Methods

Plant materials and growth conditions

GL-3 (selected from progeny of Royal Gala) plants were used for genetic transformation (Dai *et al.*, 2013). Tissue-cultured GL-3

plants for 1 month were used for apple stable transformation as previously reported (Xie *et al.*, 2018). Tissue-cultured GL-3 plants were sub-cultured every 4 wk on Murashige & Skoog (MS) medium (4.43 g l⁻¹ MS salts, 0.2 mg l⁻¹ IAA, 7.5 g l⁻¹ agar, 0.2 mg l⁻¹ 6-BA, 30 g l⁻¹ sucrose, pH 6.0). Then 4-wk-old transgenic plants and GL-3 were transferred on rooting medium (2.215 g l⁻¹ MS salts, 0.5 mg l⁻¹ IAA, 7.5 g l⁻¹ agar, 0.5 mg l⁻¹ IBA, 20 g l⁻¹ sucrose, pH 6.0) for 2 months. The plants were then transplanted into plant culture substrates (Pindstrup, Ryomgaard, Denmark) in a growth chamber with long-day conditions (16 h : 8 h, light : dark at 25°C).

To detect the expression pattern of *MdZAT5*, 8-wk-old apple (*Malus sieversii* (Led.) Roem) plants were cultivated hydroponically for an additional month and treated with 20% PEG 8000 for 6 h (Geng *et al.*, 2019). In addition, 2-month-old apple (*Malus domestica* Borkh cv. 'Golden delicious') plants were transferred to a glasshouse and cultivated for an additional month before drought treatment irrigation was withheld for up to 8 d.

To investigate the tissue-specific expression of *MdZAT5*, newly formed roots, stems from newly produced shoots, mature leaves were collected from 5-yr-old apple (*Malus prunifolia* (Willd.) Borkh) trees.

'Orin' (*M. domestica* Borkh) calli were grown on MS media (4.43 g l⁻¹ MS salts, 8 g l⁻¹ agar, 1.5 mg l⁻¹ 2,4-D, 0.4 mg l⁻¹ 6-BA, 30 g l⁻¹ sucrose, pH 5.8) in darkness at 24°C.

Generation of transgenic apple plants

To generate transgenic apple plants, the coding sequence (CDS) of *MdZAT5* was cloned and introduced into the pK2GW7 and a 205-bp fragment of the *MdZAT5* CDS was cloned into pK7GWIWG2D vector based on gateway recombination technology (Invitrogen). The vector pK2GW7 was used as an overexpression vector and the pK7GWIWG2D vector was used as the RNA silencing vector. *Agrobacterium*-mediated transformation of apple was performed as previously described (Dai *et al.*, 2013). GL-3 was used as the genetic background and *Agrobacterium* strain EHA105 was used. The primers used for constructing these vectors are shown in Table S1.

Stem-loop quantitative polymerase chain reaction

Stem-loop quantitative polymerase chain reaction (qPCR) was performed as described by Niu *et al.* (2019). The primers used for this study are listed in Table S1. MDH was used as reference gene.

Chromatin immunoprecipitation qPCR

Chromatin immunoprecipitation qPCR (ChIP-qPCR) assay was performed as described by Xie *et al.* (2018). Briefly, the leaves of GL-3 plants were crosslinked in 1% (w/v) formaldehyde and chromatin were isolated and sonicated. Then, immunoprecipitation was performed with or without polyclonal antibody against *MdZAT5* produced from rabbit (GenScript, Nanjing, China) overnight at 4°C. The chromatin antibody complex was

precipitated with ChIP-grade protein A/G agarose beads (Thermo Scientific, Waltham, MA, USA) and washed with different washing buffers four times. Finally, after reversing cross-link and protein removal, recovered DNA was used for qPCR. Primers used for ChIP-qPCR are listed in Table S1. MDH was used as reference gene.

Electrophoretic mobility shift assay

Truncated MdZAT5 (from aa 1 to aa 233) was cloned into vector pGEX4T and expressed in the *Escherichia coli* strain BL21 (DE3) with 0.6 mM isopropylthio- β -galactoside for 20 h at 18°C. The glutathione S-transferase (GST)-tagged protein was purified using PierceTM Glutathione Spin Columns (Thermo ScientificTM, Rockford, IL, USA). The purity of the proteins was analyzed by a 10% sodium dodecyl sulfate–polyacrylamide gel electrophoresis (SDS-PAGE) gel stained with Coomassie Brilliant Blue R-250. The electrophoretic mobility shift assay (EMSA) was performed according to the manual of LightShift Chemiluminescent EMSA Kit (89880; Thermo Scientific). The oligonucleotide probes labeled with biotin are listed in Table S1.

Accession numbers

Sequence data from this article can be found in GDR under the following accession nos.: MdZAT5 (MDP0000769354), MdHYL1 (MD15G1387000), MdATG8a (MD10G1319000), MdLOX3 (MD07G1004600); and in NCBI under the following nos.: MdNRT1.5 (XP_008373542.2), MdRHA2a (XP_008389249.1), MdTPL2 (XP_008390610.1), MdNAS3 (XP_008370376.1), MdXTH5 (NP_001315789.1), MdABCG11 (XP_008388513.1), MdSEOB (XP_008374367.2), MdLecrk1.1 (XP_008338483.2), MdLEA14 (XP_008344736.1), MdSPMS (XP_028962045.1), MdTPX1 (XP_008386744.1) and MdCAT3 (XP_008348924.2).

Results

Identification of the apple drought-responsive gene *MdZAT5*

We previously performed an RNA-Seq analysis *Malus sieversii* and identified one C2H2 ZFP that was induced by PEG (Geng *et al.*, 2019). To understand the nature of this protein, we cloned this gene (MDP0000769354) from *M. × domestica* cv. Golden Delicious. Protein sequence analysis showed that it contained two conserved C2H2 domains at its N terminus and an ethylene-responsive element binding-factor-associated amphiphilic repression (EAR) motif at its C terminus (Fig. S1a). Protein sequence alignment showed that it was homologous to *Arabidopsis* ZAT5, we therefore named it *MdZAT5*. We next examined the induction of *MdZAT5* in response to PEG treatment in GL-3 (a progeny of Royal Gala) plants using reverse transcription qPCR (RT-qPCR). Results showed that *MdZAT5* was indeed induced (c. 2.5-fold) upon PEG treatment (Fig. 1a), and *MdZAT5* was also upregulated under short-term drought treatment (Fig. 1b).

These results indicated that MdZAT5 might play a role in the apple drought response.

A phylogenetic tree of ZAT5 proteins from 15 different plant species revealed that all proteins contained two conserved C2H2 domains and an EAR motif, and MdZAT5 was highly homologous to PbZAT5 from *Pyrus bretschneideri* (Fig. S1b). In addition, tissue-specific expression analysis showed that *MdZAT5* was predominantly expressed in the stem and leaves (Fig. 1c). As expected, MdZAT5 was localized in the nucleus (Fig. 1d). Yeast two-hybrid (Y2H) assay showed that MdZAT5 did not have a self-activation activity (Fig. 1e).

MdZAT5 is a positive regulator of drought stress tolerance

To investigate the function of *MdZAT5* in drought stress tolerance, we first produced *MdZAT5* transgenic apple plants. We used an RNA interference (RNAi) approach with GL-3 as the genetic background to generate transgenic RNAi apple plants. In the two independent transgenic lines (*MdZAT5* RNAi #8 and #12), *MdZAT5* expression was reduced c. 60–90% (Fig. S2). We also generated *35S:MdZAT5*-overexpressing transgenic plants (*MdZAT5* OE #6 and #22) in which *MdZAT5* expression was increased 6–7-fold (Fig. S2).

We then applied short-term drought stress to the *MdZAT5* RNAi and overexpression (OE) plants. The leaves of the RNAi plants wilted faster than those of GL-3 under drought stress. After 25 d of drought, the leaves of RNAi plants were drier and more brittle than the GL-3 leaves. After 30 d of drought, the GL-3 leaves had wilted completely, whereas the leaves of the OE plants performed better (Fig. 2a,b). We also recorded the survival ability of the transgenic and control plants under short-term drought stress. After 25 d of drought followed by 8 d of recovery, the survival percentage was lower in *MdZAT5* RNAi plants compared with GL-3 plants. However, after 30 d of drought, the survival percentage of *MdZAT5* OE plants was higher than that of GL-3 plants. As shown in Fig. 2(c,d), only 15–18% of the *MdZAT5* RNAi plants survived, compared with 40% of the GL-3 plants and 50–55% of the *MdZAT5* OE plants. Moreover, compared with the GL-3 plants, the *MdZAT5* RNAi plants showed higher electrolyte leakage after 7 d of drought exposure, whereas the *MdZAT5* OE plants showed lower electrolyte leakage (Fig. 2e,f). These results suggest that MdZAT5 is a positive regulator of apple drought tolerance.

MdZAT5 regulates expression of drought-responsive genes under drought

To further determine how MdZAT5 functions in the drought stress response and identify its potential downstream targets under drought, we performed an RNA-Seq analysis with GL-3 and *MdZAT5* RNAi plants under simulated drought treatment (Table S2; Fig. S3). Relevant methods are included in Methods S1. Whole-genome expression analysis showed that 2126 genes were induced and 2526 genes were repressed in GL-3 leaves after simulated drought (Table S3). Compared with GL-3 plants, 1637 genes were upregulated and 643 genes were downregulated

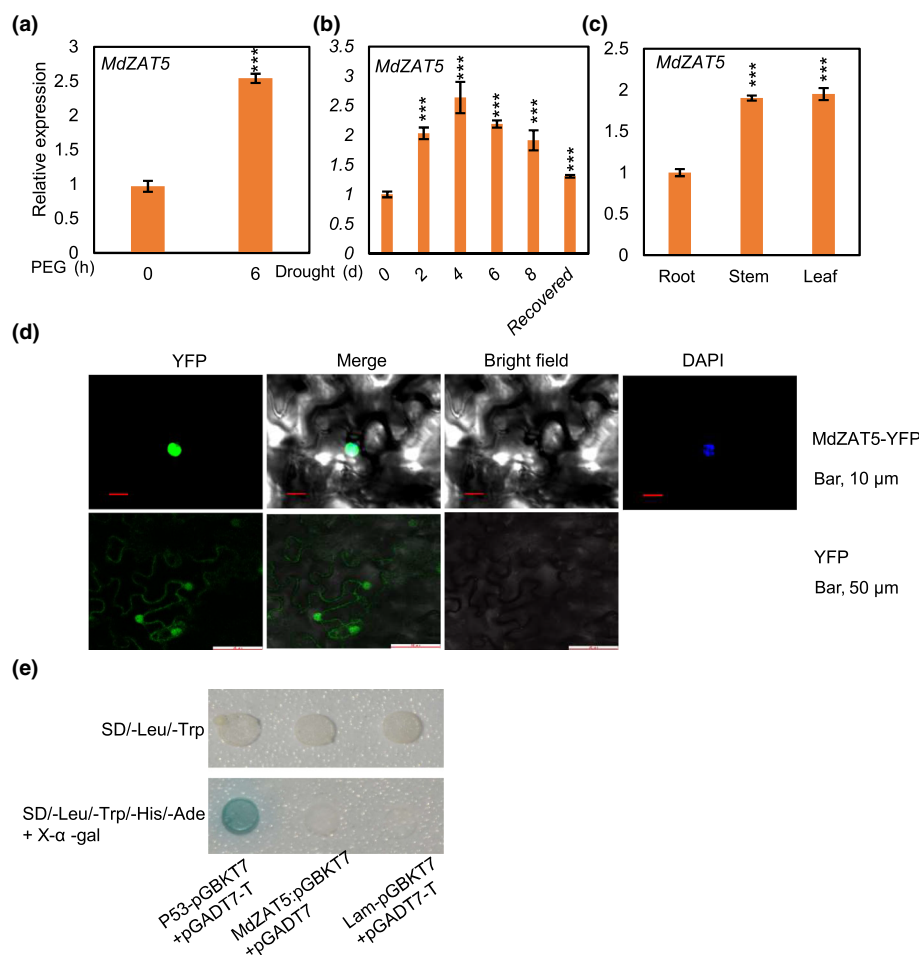


Fig. 1 *MdZAT5* expression and protein localization. (a) *MdZAT5* expression in response to PEG treatment. Two-month-old apple (*Malus sieversii*) plants were cultivated hydroponically for one additional month and treated with 20% PEG 8000 for 6 h. (b) *MdZAT5* expression in response to drought stress. Two-month-old apple (*Malus domestica*) plants grown in a glasshouse were treated with drought for 0, 2, 4, 6, 8 d, and recovered with water. (c) Tissue-specific expression of *MdZAT5* in *Malus prunifolia*. (d) *MdZAT5* localization. YFP, yellow fluorescent protein. The fluorescence signals of YFP and 4',6-diamidino-2-phenylindole (DAPI) were detected by dual-channel confocal microscopy. DAPI was used to detect nuclei. Bars: 10 μ m for *MdZAT5*-YFP; 50 μ m for YFP. (e) Transcription activity of *MdZAT5*. Yeast cells were grown on either SD/-Leu/-Trp or SD/-Leu/-Trp/-His/-Ade + X- α -gal media. Error bars indicate standard deviation (n = 3). Student's two-tailed t-test was performed and statistically significant differences were indicated by ***, $P < 0.001$.

in *MdZAT5* RNAi plants under control conditions (Table S4). Under simulated drought, 325 genes were differentially expressed in *MdZAT5* RNAi plants relative to GL-3 plants: 161 downregulated and 164 upregulated (Table S5). Among these 325 differentially expressed genes (DEGs), 54 genes that were induced by drought were downregulated in *MdZAT5* RNAi plants under drought stress, 18 genes that were repressed by drought were upregulated in *MdZAT5* RNAi plants, suggesting that *MdZAT5* regulates the expression of drought-responsive genes (Fig. 3a). A Gene Ontology (GO) enrichment analysis revealed that all DEGs in drought-treated *MdZAT5* RNAi plants compared with GL-3 plants under drought conditions were enriched in biological processes related to stimulus response and metabolic processes, particularly stress (Fig. S4). Moreover, H_2O_2 and ROS response pathways were also significantly enriched (Fig. S4).

Next, we verified the RNA-Seq expression patterns of 10 genes using RT-qPCR. We selected homologs of the following *Arabidopsis* genes: *RING-H2 FINGER A2A* (*RHA2A*), *LIPOXYGENASE 3* (*LOX3*), *L-TYPE LECTIN RECEPTOR KINASE IX.1* (*LECRK-IX.1*), *SIEVE ELEMENT OCCLUSION B* (*SEOB*), *ATP-BINDING CASSETTE G11* (*ABCG11*), *SPERMINE SYNTHASE* (*SPMS*), *AUTOPHAGY 8A* (*ATG8A*), *NITRATE TRANSPORTER 1.5* (*NRT1.5*), *XYLOGLUCAN ENDOTRANSGLUCOSYLASE/HYDROLASE 5* (*XTH5*), and *NICOTIANAMINE SYNTHASE 3* (*NAS3*) (Figs 3b, S5). *RHA2A*, *LOX3*, and *ATG8A*

are positive regulators of drought tolerance in *Arabidopsis*, rice or *Medicago*, whereas *NRT1.5* is a negative regulator (Liu *et al.*, 2008; Bu *et al.*, 2009; Li *et al.*, 2011, 2020; Chen *et al.*, 2012). RT-qPCR analysis showed that *MdZAT5* positively regulated the expression of *MdRHA2A*, *MdLOX3*, *MdLECRK-IX.1*, *MdSEOB*, *MdABCG11*, *MdSPMS*, and *MdATG8A* and negatively regulated the expression of *MdNRT1.5*, *MdXTH5*, and *MdNAS3* under PEG treatment, consistent with the RNA-Seq data (Fig. 3b; Table S5). We calculated the correlation between RT-qPCR and RNA-Seq data, and the results showed that RT-qPCR and RNA-Seq data had significant correlation (Fig. S5). We also detected the expression of these genes (shown in Fig. 3) in *MdZAT5* OE plants using RT-qPCR (Fig. S6). The results were consistent with RNA-Seq data using *MdZAT5* RNAi plants. These results confirmed the reliability of the RNA-Seq results and suggested that *MdZAT5* may promote apple drought tolerance by regulating the expression of drought-responsive genes.

Identification of *MdZAT5* binding sites and potential direct targets

Transcription factors usually bind to a specific *cis*-element in the promoters of their targets (Singh *et al.*, 2002). To gain deeper insight into how *MdZAT5* regulates the expression of drought-responsive genes, we performed DNA affinity purification and

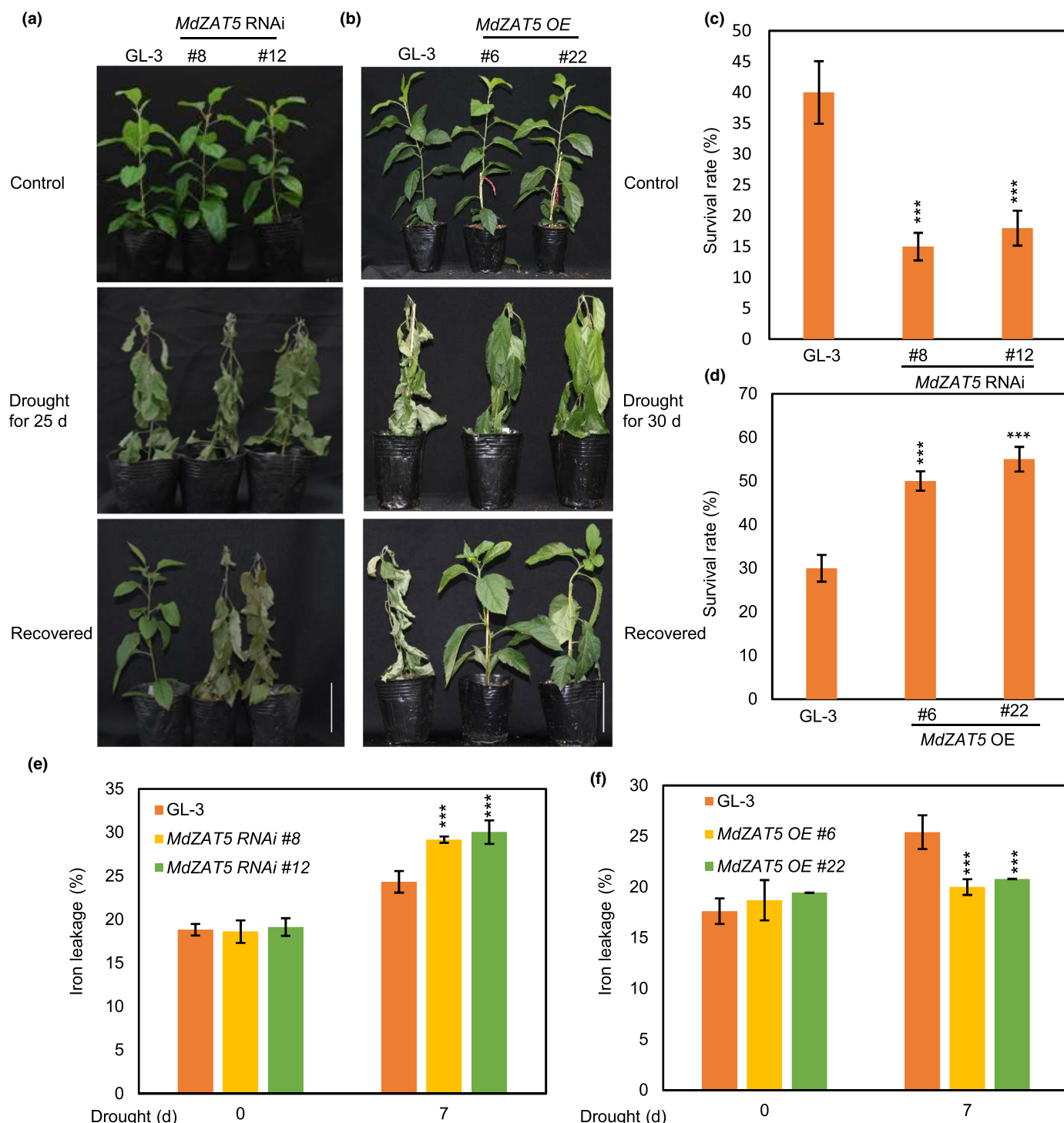


Fig. 2 MdZAT5 is a positive regulator of the apple drought response. (a) Drought tolerance of GL-3 and transgenic *MdZAT5* RNA interference (RNAi) plants. Two-month-old plants were treated with drought for 25 d, and then reirrigated for 8 d. (b) Drought tolerance of GL-3 and transgenic *MdZAT5* overexpressed (OE) plants. Two-month-old plants were treated with drought for 30 d, and then reirrigated for 8 d. (c) Survival rate of GL-3 and transgenic *MdZAT5* RNAi plants under drought stress. (d) Survival rate of GL-3 and transgenic *MdZAT5* OE plants under drought stress. (e) The ion leakage of GL-3 and transgenic *MdZAT5* RNAi plants under drought stress. (f) The ion leakage of GL-3 and transgenic *MdZAT5* OE plants under drought stress. Bars: (a, b) 10 cm. Error bars indicate standard deviation ($n = 21$ in (c, d) and $n = 5$ in (e, f)). Student's two tailed *t*-test was performed and statistically significant differences were indicated by ***, $P < 0.001$.

sequencing (DAP-Seq) using MdZAT5-GST protein (Fig. S7). After analyzing the sequenced reads, we identified > 9000 potential MdZAT5 targets (Table S6). Over 73% of the identified peaks

were located in promoter regions, within 2 kb upstream of the transcription start site (Fig. 4a). Less than 13% of the identified peaks were located in introns and exons (Fig. 4a). The peaks were

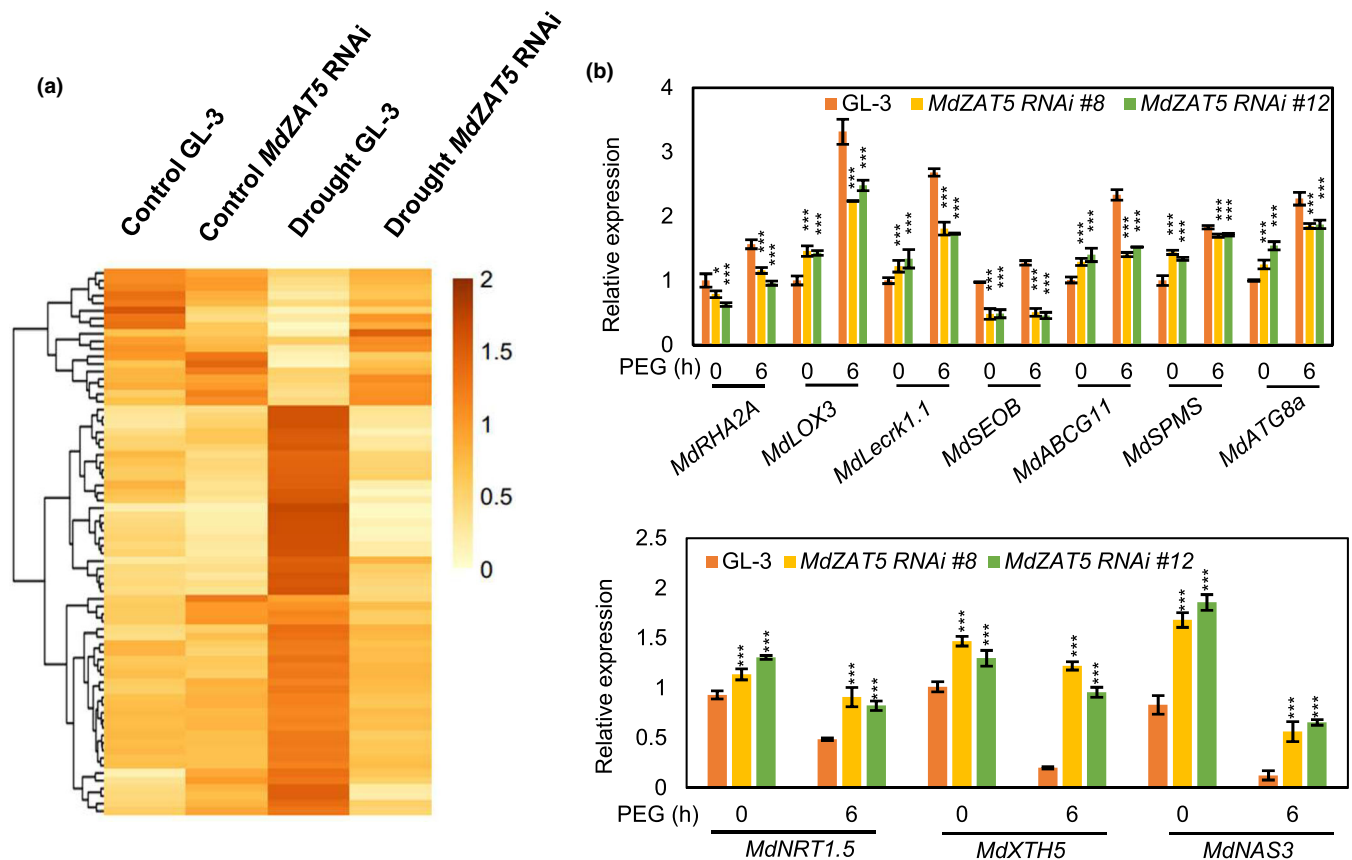


Fig. 3 Genome-wide gene expression profiling in *MdZAT5* RNA interference (RNAi) and GL-3 plants in response to simulated drought stress. (a) Heatmap analysis of differentially expressed genes (DEGs) in *MdZAT5* RNAi and GL-3 plants in response to simulated drought stress. Leaves detached from 2-month-old *Malus domestica* plants which were cultivated hydroponically for one additional month and treated with 20% PEG 8000 for 6 h. Heatmap represents the transcription levels. (b) Verification of RNA sequencing (RNA-Seq) data. Error bars indicate standard deviation ($n = 3$). Student's two tailed t -test was performed and statistically significant differences were indicated by *, $P < 0.05$ or ***, $P < 0.001$.

highly enriched from -500 to -250 bp from the transcription start site, TSS (Fig. 4b). Gene Ontology term enrichment performed on the potential *MdZAT5* targets indicated that they were mainly associated with response to abiotic and biotic stresses, such as salt, drought, cold, defense against fungus or virus, and hormone transport (Fig. S8). Furthermore, some pathways essential for plant growth and development were also enriched, including leaf development, plant organ senescence, and chromatin organization (Fig. S8), implying that *MdZAT5* has diverse functions.

Using HOMER, we identified one distinct consensus sequence for *MdZAT5* binding, T/ACACT/AC/A/G (Fig. 4c). Yeast one-hybrid (Y1H) assays confirmed the binding of *MdZAT5* to TCACTC, TCACTA, TCACTG, ACACTC, ACACTA, ACACTG, TCACAC, TCACAG, TCACAA, and ACACAG motifs. Y1H assays using ACACAC and ACACAA *cis*-elements were not successful because both elements could not be repressed by aureobasidin A (AbA) (Fig. 4d). We also performed EMSA using a *MdZAT5*-GST fusion protein in which the N terminal region of *MdZAT5* that contained the C2H2 domain (1–233 aa) was fused with GST to determine whether *MdZAT5* could bind to the dominant DNA element TCACTC. Electrophoretic mobility shift assay analysis showed a clear interaction between *MdZAT5* and a labeled probe that repeated TCACTC six times (Fig. 4e).

Among the 9000 potential target genes, 798 genes were responsive to drought stress (Fig. 4f). One of the potential target genes was a homolog of *RHA2A*, a positive regulator of drought stress tolerance in *Arabidopsis* (Bu *et al.*, 2009; Li *et al.*, 2011) (Table S6; Fig. S9). We analyzed the promoter sequence of *MdRHA2A* and found that it contained six *MdZAT5* recognition *cis*-elements, one TCACTC, one ACACTA, one ACACTC, one TCACTA, and two TCACAA motifs (Fig. S10). We performed a ChIP-qPCR assay to determine whether *MdZAT5* could directly bind to these motifs. The results showed that *MdZAT5* bound to fragment a in the *MdRHA2A* promoter (-1539 to -1275 bp) (Fig. 5a,b). We subsequently performed an EMSA, and found that *MdZAT5* directly bound to the motifs of fragment a in the *MdRHA2A* promoter (Fig. 5c). Another potential *MdZAT5* target was a homolog of *Arabidopsis* *LEA14*, a stress-related marker gene (Ma *et al.*, 2018; Table S6; Fig. S11). We analyzed the promoter sequence of *MdLEA14* and found that it contained 10 *MdZAT5* recognition *cis*-elements (Fig. S12). We then performed a ChIP-qPCR assay and found that *MdZAT5* bound to fragment a in the *MdLEA14* promoter (-958 to -761 bp) (Fig. 5d,e). Electrophoretic mobility shift assay results confirmed that *MdZAT5* directly bound to the motifs of fragment a in the *MdLEA14* promoter (Fig. 5f). Moreover,

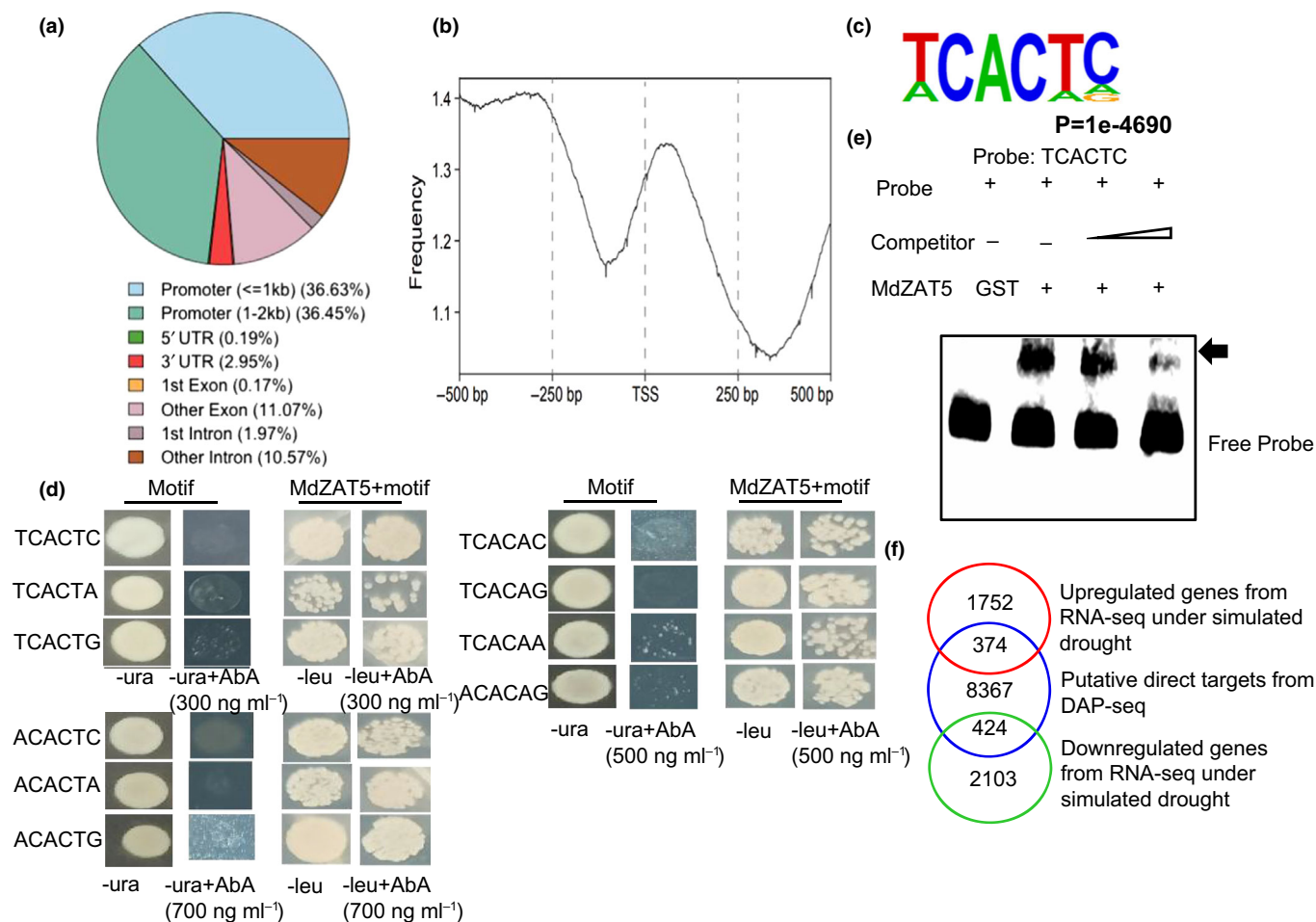


Fig. 4 DNA affinity purification and sequencing (DAP-Seq) analysis and MdZAT5 recognition sites. (a) Distribution of MdZAT5 binding regions in the apple genome based on DAP-Seq analysis. (b) Peak distance from the TSS. (c) A potential recognition motif of MdZAT5 revealed by DAP-Seq. (d) Yeast-one hybrid analysis of MdZAT5 and its binding sites. (e) Electrophoretic mobility shift assay (EMSA) analysis. The N terminal region of MdZAT5 containing the C2H2 domain (1–233 aa) was used. Arrows indicate the protein–DNA complex. (f) Numbers of genes detected by DAP-Seq and RNA-seq under simulated drought conditions. Differentially expressed genes (DEGs) were detected by RNA-Seq in GL-3 plants in response to simulated drought conditions. Recombinant MdZAT5 protein was used to perform DAP-Seq.

MdRHA2A and *MdLEA14* were induced by drought stress, and their expression was downregulated in *MdZAT5* RNAi plants and upregulated in *MdZAT5* OE plants under control and short-term drought conditions (Fig. 5g,h). We further performed dual-luciferase (LUC) assays with the LUC reporter gene driven by the *MdRHA2A* or *MdLEA14* promoter. LUC activity analysis suggested that MdZAT5 could directly activate expression of *MdRHA2A* and *MdLEA14* under normal or dehydration conditions (Fig. 5i,j). These results suggested that MdZAT5 may regulate apple drought tolerance by modulating the expression of the drought-responsive genes, such as *MdRHA2A* and *MdLEA14*.

MdZAT5 stimulates H₂O₂ scavenging under drought conditions by binding to the promoters of the antioxidase genes

The close relationship between ROS and plant drought tolerance is well established. Previous studies have reported that levels of ROS increase under drought stress, and the accumulation of

ROS is higher in drought-sensitive plants than in drought-tolerant plants (Zou *et al.*, 2015; Choudhury *et al.*, 2017). Gene Ontology enrichment analysis of the DEGs regulated by MdZAT5 suggested the involvement of H₂O₂ and ROS response pathways in MdZAT5-mediated drought tolerance. This finding prompted us to examine H₂O₂ accumulation in the leaves of GL-3 and *MdZAT5* transgenic plants after short-term drought treatment. Plants that overexpressed *MdZAT5* accumulated less H₂O₂ under drought stress than GL-3 plants, whereas *MdZAT5* RNAi plants accumulated more H₂O₂ (Fig. 6a,d). In plants, H₂O₂ is scavenged by POD and CAT. We found that the activity of POD and CAT was higher in *MdZAT5*-overexpressing plants but lower in *MdZAT5* RNAi plants than in GL-3 plants under short-term drought stress (Fig. 6b–f). Consistently, the expression of *MdCAT3*, *MdTPX1*, and *MdCAT2* (homologs of *AtCAT3*, *THIOREDOXIN-DEPENDENT PEROXIDASE 1* (*AtTPX1*), and *AtCAT2*, respectively) was upregulated in *MdZAT5*-overexpressing plants but downregulated in *MdZAT5* RNAi plants under short-term drought stress (Fig. 6g–i),

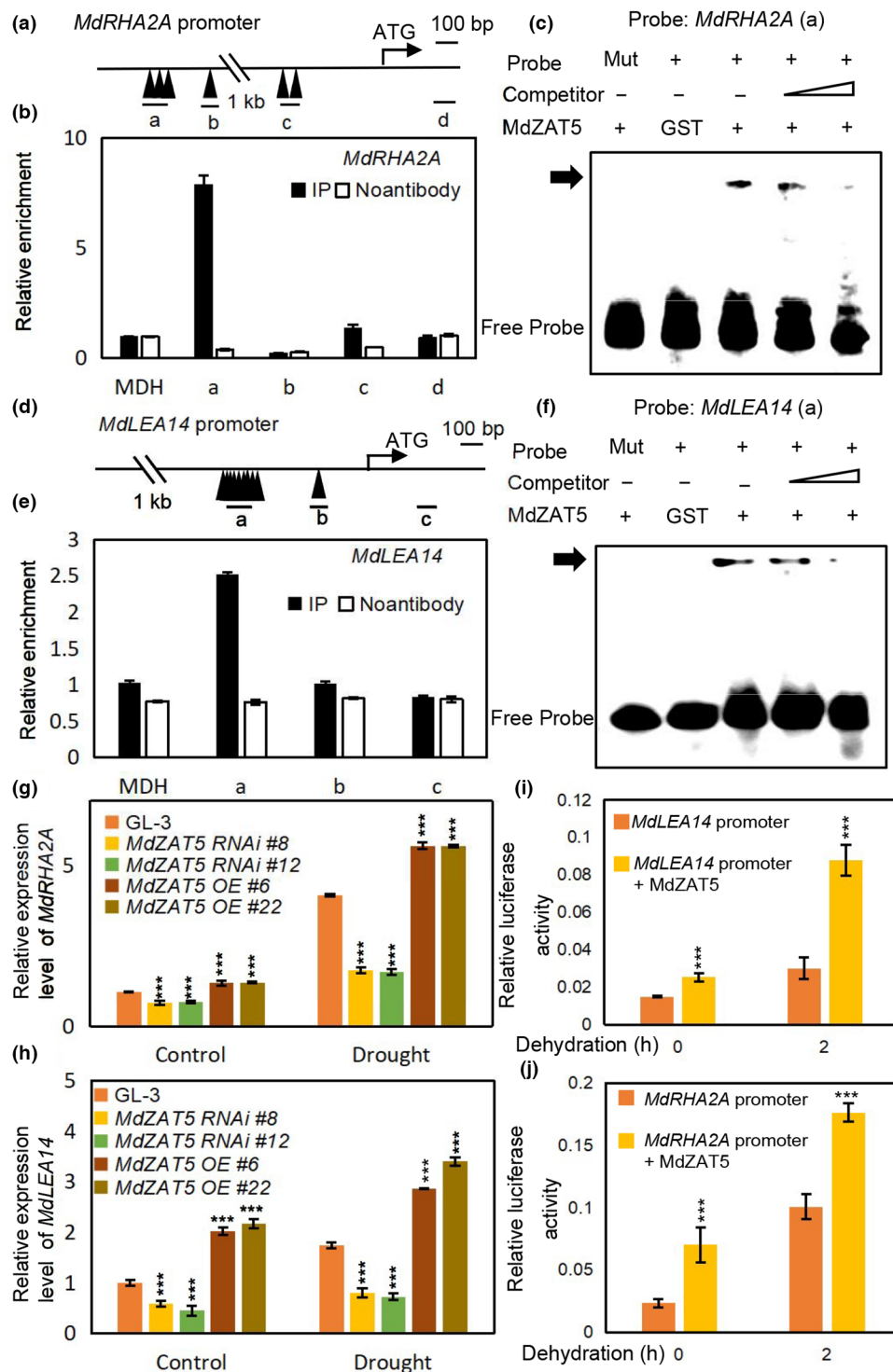


Fig. 5 MdZAT5 binds to the promoter regions of drought-responsive genes. (a) Diagram of *MdRHA2A* promoter regions. Fragment a contains one TCACCTC, one ACACTA, and one TCACCTA motif; fragment b contains one TCACAA motif; fragment c contains one ACACTC and one TCACAA motif; fragment d serves as a negative control. (b) Binding of MdZAT5 to the promoter of *MdRHA2A* by chromatin immunoprecipitation quantitative polymerase chain reaction (ChIP-qPCR) analysis. Fragment d, MDH, and no antibody serve as negative controls. (c) Binding of MdZAT5 to the promoter of *MdRHA2A* by electrophoretic mobility shift assay (EMSA) analysis. (d) Diagram of *MdLEA14* promoter regions. Fragment a contains four TCACCTC, two TCACAC, one TCACAA, one ACACTC, and one TCACCTA motif; fragment b contains one TCACAA motif; fragment c serves as a negative control. (e) ChIP-qPCR analysis of MdZAT5 bound to the promoter of *MdLEA14*. Fragment c, MDH, and no antibody serve as negative controls. (f) EMSA analysis of MdZAT5 bound to the promoter of *MdLEA14*. (g, h) Relative messenger RNA (mRNA) level of *MdRHA2A* and *MdLEA14* in *Malus domestica* under control or drought treatment for 7 d. (i, j) MdZAT5 can bind to *MdLEA14* and *MdRHA2A* promoters by dual-luciferase assay. Pro35S::REN was used as an internal control. Quantification was performed by normalizing firefly luciferase activity to that of renilla luciferase. Mut represents the mutant probe. Binding sites in mutant probe were replaced with AAAAAA. Arrows indicate protein–DNA complex. Error bars indicate standard deviation ($n = 5$ in (i) and (j), $n = 3$ in (b), (e), (g), and (h)). Student's two tailed t -test was performed and statistically significant differences were indicated by ***, $P < 0.001$.

indicating that MdZAT5 promotes H_2O_2 scavenging by modulating the expression and activities of antioxidant enzymes.

To examine whether MdZAT5 could bind directly to the promoters of these genes, we analyzed their promoter sequences and found that *MdTPX1* and *MdCAT3* promoters contained MdZAT5 binding sites (Figs S13–S16). We then performed ChIP-qPCR assays and found that MdZAT5 bound to fragment a in the *MdTPX1* promoter (–622 to –470 bp) and fragment a

in the *MdCAT3* promoter (–1832 to –1664 bp) (Fig. 6j–m). We subsequently performed EMSAs and confirmed that MdZAT5 bound directly to motifs of fragment a in the *MdTPX1* and *MdCAT3* promoters (Fig. 6n,o). Dual-LUC assays suggested that MdZAT5 could directly activate expression of *MdTPX1* and *MdCAT3* under dehydration conditions (Fig. 6p,q), which was consistent with RT-qPCR results (Fig. 6g,i). Overall, these results suggested that MdZAT5 stimulates H_2O_2 scavenging under

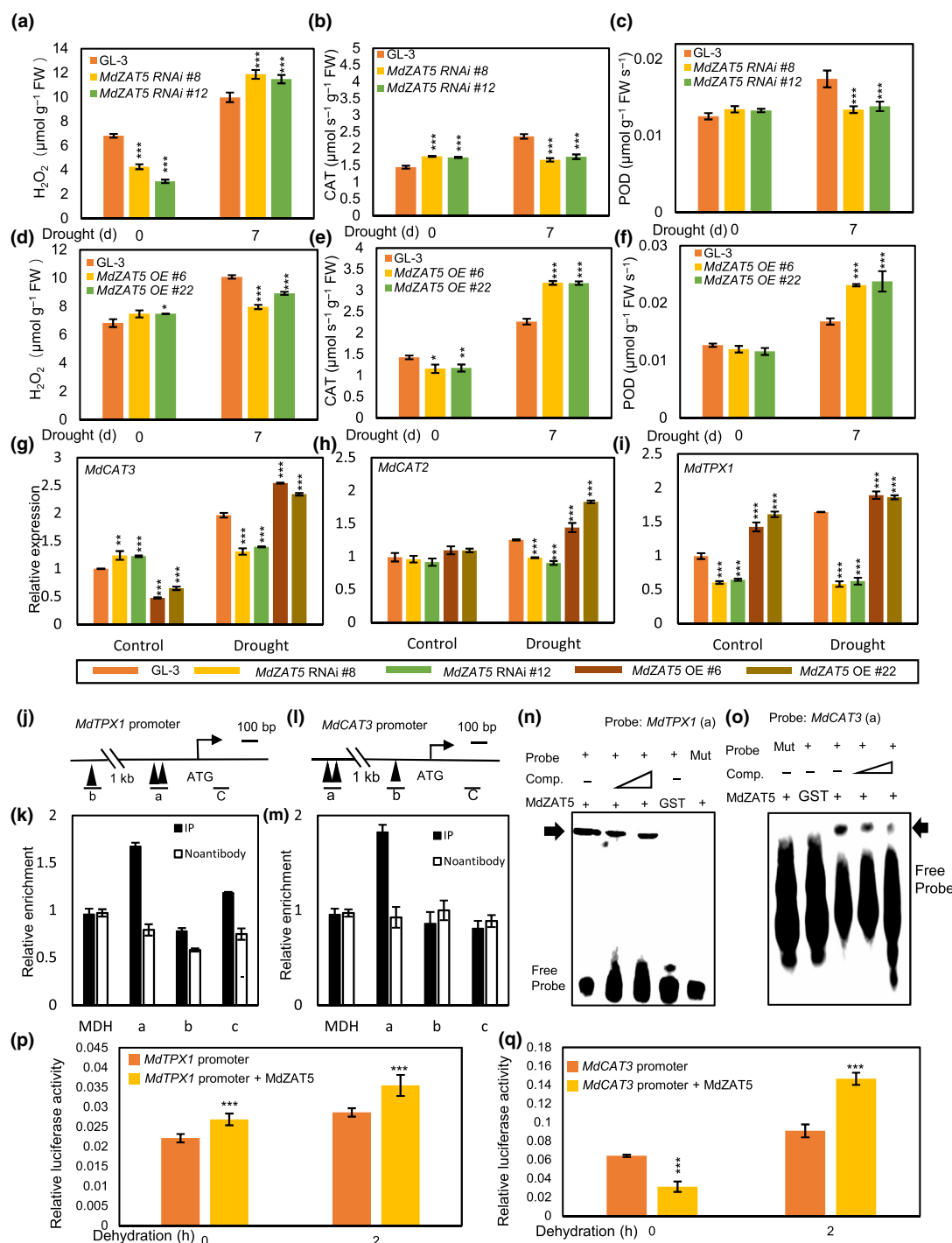


Fig. 6 MdZAT5 regulates reactive oxygen species (ROS) homeostasis under drought stress by binding to the promoter regions of antioxidant genes. (a–c) The hydrogen peroxide (H₂O₂) content (a), and activities of catalase (CAT) (b) and peroxidase (POD) (c) of GL-3 and transgenic MdZAT5 RNA interference (RNAi) plants under control and drought conditions. (d–f) The H₂O₂ content (d), and activities of CAT (e) and POD (f) of GL-3 and transgenic MdZAT5 overexpressed (OE) plants under control and drought conditions. Error bars indicate standard deviation ($n = 5$). (g–i) Relative messenger RNA (mRNA) level of MdCAT3, MdCAT2, and MdTPX1 in *Malus domestica* under control or drought treatment for 7 d. (j) Diagram of MdTPX1 promoter. Fragment a contains one TCACAC and one TCACTG motif; fragment b contains one TCACTC motif. Fragment c serves as a negative control. (k) Chromatin immunoprecipitation quantitative polymerase chain reaction (ChIP-qPCR) analysis of MdZAT5 bound to the promoter of MdTPX1. Fragment c, MDH, and no antibody serve as negative controls. (l) Diagram of MdCAT3 promoter. Fragment a contains one TCACTA and one TCACTG motif; fragment b contains one TCACAA motif. Fragment c serves as a negative control. (m) ChIP-qPCR analysis of MdZAT5 bound to the promoter of MdCAT3. Fragment c, MDH, and no antibody serve as negative controls. (n, o) Electrophoretic mobility shift assay (EMSA) analysis of MdZAT5 bound to the promoters of MdTPX1 and MdCAT3. (p, q) MdZAT5 can bind to MdTPX1 and MdCAT3 promoters by dual-luciferase assay. Mut represents the mutant probe. Binding sites in mutant probe were replaced with AAAAAA. Arrows indicate protein–DNA complex. Error bars indicate standard deviation ($n = 5$ in (a–f), (p), and (q), $n = 3$ in (g), (h), (i), (k) and (m)). Student's two tailed *t*-test was performed and statistically significant differences were indicated by *, $P < 0.05$; **, $P < 0.01$ or ***, $P < 0.001$.

drought conditions by promoting the expression of these antioxidant enzyme encoding genes.

MdHYL1 interacts with MdZAT5 and is a target of MdZAT5

To further examine the molecular mechanisms by which MdZAT5 functions in response to drought stress, we performed a Y2H screening to identify potential MdZAT5 interacting proteins and identified the dsRNA binding protein, MdHYL1 (Figs S17, S18). In Y2H assay, the yeast cells co-transformed with MdZAT5-pGBKT7 and MdHYL1-pGADT7 plasmids can grow normally on SD/-Leu/-Trp/-His/-Ade/X- α -gal, indicating that MdZAT5 interacts with MdHYL1 (Fig. 7a). We also performed split-LUC assay to further verify the interaction between MdHYL1 and MdZAT5 *in vivo*, as LUC activity was observed only when MdHYL1-cLUC (C-terminal half of luciferase) co-infiltrated with MdZAT5-nLUC (N-terminal half of luciferase) (Fig. 7b). Moreover, a co-immunoprecipitation (Co-IP) analysis further confirmed MdHYL1 could interact with MdZAT5 *in planta* (Fig. 7c). Furthermore, a co-localization assay showed that MdHYL1 could be co-localized with MdZAT5 in the nucleus (Fig. 7d).

Interestingly, we found the MdZAT5 binding *cis*-elements in the *MdHYL1* promoter (Fig. S19). Using EMSA and ChIP-qPCR analyses, we confirmed that MdZAT5 could bind to the promoter region of *MdHYL1* (Fig. 7e,f). ChIP-qPCR results showed that MdZAT5 bound to the fragments a, b, c, and e of *MdHYL1* promoter under normal conditions while bound to the fragments d and e under drought (Fig. 7e). In addition, we found that under control conditions, *MdHYL1* was negatively regulated by MdZAT5 while it was positively regulated by MdZAT5 under drought conditions by RT-qPCR and Dual-LUC assays (Fig. 7g, h), which may due to the different binding sites of MdZAT5 in the promoter of *MdHYL1* under normal or drought conditions (Fig. 7e).

MdZAT5 affects miRNA biogenesis under control and drought conditions

In apple, MdHYL1 positively regulates drought tolerance by mediating abundance of drought-responsive miRNAs (Shen *et al.*, 2022). The interaction between MdHYL1 and MdZAT5 prompted us to examine whether MdZAT5 functioned in the miRNA biogenesis pathway. Stem-loop qPCR analysis showed that all tested miRNAs (Mdm-miR164, Mdm-miR398, Mdm-miR408, Mdm-miRn-249, Mdm-miR2111, Mdm-miR156, Mdm-miR7121, Mdm-miR395, Mdm-miR396, Mdm-miR166, Mdm-miR160, Mdm-miR172, and Mdm-miR171i) were significantly more abundant in *MdZAT5* RNAi plants than in GL-3 plants (Fig. 8a), indicating a negative role of MdZAT5 in miRNA biogenesis under control conditions. Northern blot assays suggested that the accumulation of Mdm-miR408, Mdm-miR172, and Mdm-miR160 in *MdZAT5* RNAi plants were more than that in GL-3 plants (Figs S20, S21), which was consistent with stem-loop qPCR results. We also tested expression of the miRNA targets in GL-3 and *MdZAT5* RNAi plants by RT-qPCR, and we

found that the expression of *MdSPL2*, *MdSPL9* (targets of miR156), *MdCSD1*, *MdCSD2* (targets of miR398), *MdARF17* (target of miR160), and *MdNAC1* (target of miR164) was clearly reduced in *MdZAT5* RNAi plants compared with GL-3 plants (Fig. 8a), consistent with higher abundance of Mdm-miR156, Mdm-miR398, Mdm-miR160, and Mdm-miR164 in *MdZAT5* RNAi plants. [Correction added after first publication 4 November 2022: in the preceding sentence miR159 was changed to miR156 in two places, and miR166 to miR398 in two places.]

We next asked whether MdZAT5 regulated miRNA biogenesis under drought stress. Stem-loop qPCR analysis of *MdZAT5* RNAi and GL-3 plants under drought stress showed that Mdm-miR160, Mdm-miR396, Mdm-miR164, Mdm-miR408, Mdm-miR156, Mdm-miRn-249, and Mdm-miR172 decreased in abundance in *MdZAT5* RNAi plants compared with GL-3 plants under drought stress, whereas Mdm-miR166, Mdm-miR171i, and Mdm-miR398 were upregulated in *MdZAT5* RNAi plants under drought stress (Fig. 8b). Northern blot assays also found that the accumulation of Mdm-miR160 in *MdZAT5* RNAi plants was less than that of in GL-3 plants under drought conditions (Fig. S21). Among these miRNAs, miR396, miR164, miR408, Mdm-miR156, Mdm-miR160, Mdm-miRn-249, and miR172 have positive roles in the drought response (Fang *et al.*, 2014; Chen *et al.*, 2015; Hajyzadeh *et al.*, 2015; Li *et al.*, 2016; Arshad *et al.*, 2017; Lu *et al.*, 2017; Niu *et al.*, 2019; Hang *et al.*, 2021; Shen *et al.*, 2022), whereas miR166, Mdm-miR171i, and miR398 have negative roles (Qin *et al.*, 2013; Zhang *et al.*, 2018; Wang *et al.*, 2020).

To further investigate how MdZAT5 affects miRNA biogenesis, we performed split-LUC assay to test the potential interaction of MdZAT5 with homologs of miRNA biogenesis factors in apple, including MdDCL1, MdSE, MdCDF2, MdNOT2, MdAGO1, and MdCDC5. The results showed that MdZAT5 interacted with MdDCL1, MdSE, and MdCDF2, but not MdNOT2, MdAGO1, and MdCDC5 *in vivo* (Fig. S22). Together, these results suggested that MdZAT5 is an important miRNA biogenesis regulator and may promote drought tolerance by regulating the expression of these drought-responsive miRNAs.

Effects of MdZAT5 on the MdHYL1-MdDCL1 interaction and abundance of MdHYL1-bound pri-miRNAs under control and drought

The interaction of MdZAT5 and MdHYL1 prompted us to test if MdZAT5 has an effect on the interaction between MdHYL1 and MdDCL1. The results showed that the interaction between MdDCL1 and MdHYL1 decreased with the addition of MdZAT5 proteins (Fig. 8c). LUC complementation imaging assays showed that when MdZAT5-GST was present, interaction between MdHYL1 and MdDCL1 was attenuated, as indicated by the decreased luminescence signal (Fig. 8d,e). However, dehydration treatment did not show obvious effects on the protein interaction (Fig. 8d,e). These results showed that MdZAT5 interrupted the interaction between MdHYL1 and MdDCL1 and that drought stress rescued the MdZAT5-mediated inhibition of the interaction between MdHYL1 and MdDCL1.

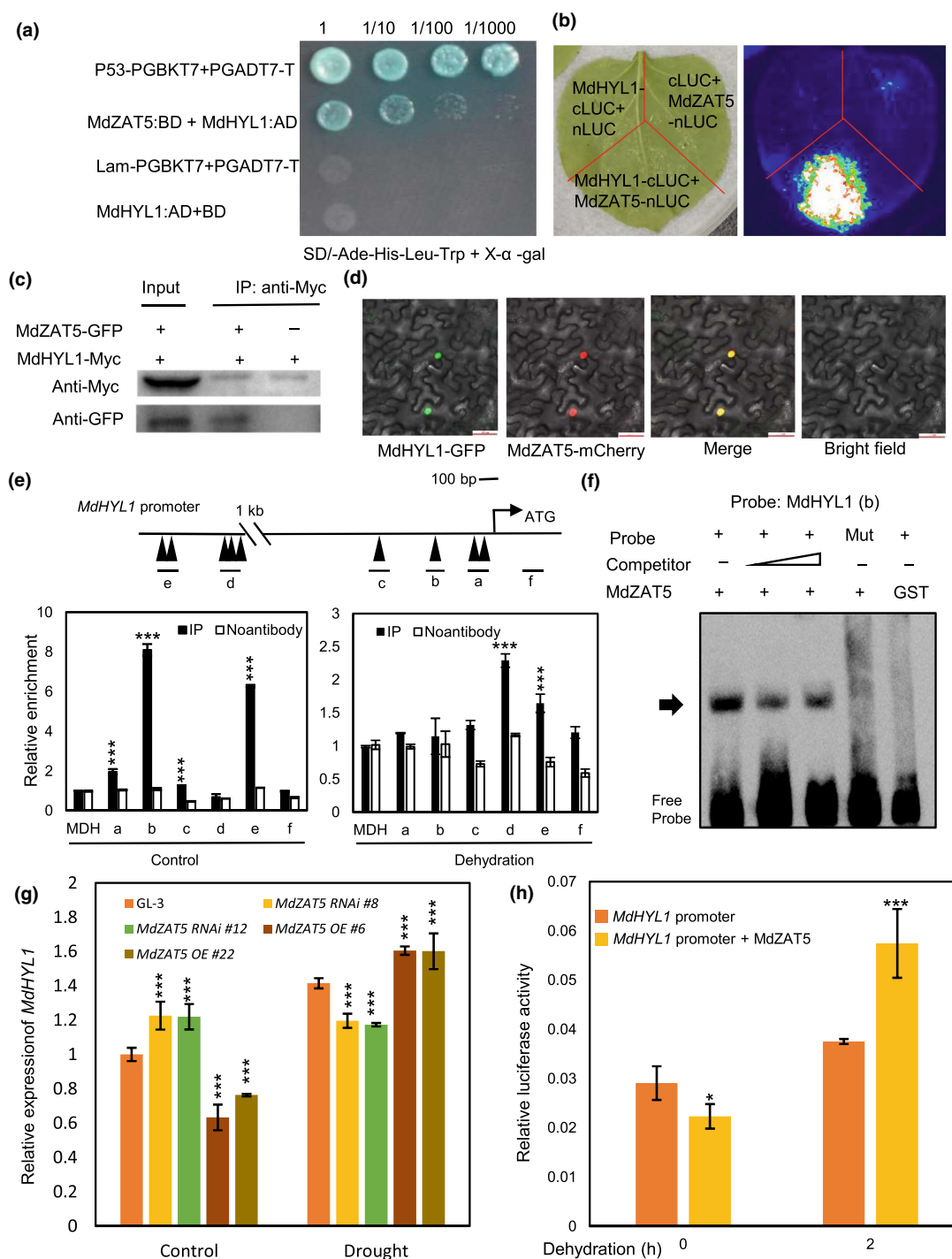


Fig. 7 MdHYL1 interacts with MdZAT5 and is a target of MdZAT5. (a) MdZAT5 interacts with MdHYL1 by yeast two-hybrid analysis. (b) MdZAT5 interacts with MdHYL1 by split luciferase analysis. (c) MdZAT5 interacts with MdHYL1 in tobacco by co-immunoprecipitation (Co-IP) analysis. MdZAT5-GFP was co-infiltrated with Myc-MdHYL1 in tobacco leaves. Proteins were extracted and immunoprecipitated with an anti-Myc antibody. The immunocomplex was then detected with Western blot using anti-Myc or anti-GFP. (d) Co-localization of MdZAT5 and MdHYL1. GFP, green fluorescent protein. mCherry, red fluorescent protein. The fluorescence signals of GFP and mCherry were detected by dual-channel confocal microscopy. Bar, 50 μ m. (e) Chromatin immunoprecipitation quantitative polymerase chain reaction (ChIP-qPCR) showing binding of MdZAT5 to the *MdHYL1* promoter under control and dehydration for 2 h. Fragment a contains one ACACAA and one AACTA motif; fragment b contains one ACACAG motif; fragment c contains one AACTA motif; fragment d contains two AACTC motifs and one TCACAA motif and fragment e contains one ACACAG and one TCACTG motif. Fragment f, MDH, and no antibody serve as negative controls. (f) Electrophoretic mobility shift assay (EMSA) showing the interaction of MdZAT5 protein and *MdHYL1* promoter. (g) The transcript level of *MdHYL1* in *Malus domestica* under control and drought conditions for 7 d. (h) MdZAT5 can bind to *MdHYL1* promoter by dual-luciferase assay. Mut represents the mutant probe. Binding sites in mutant probe were replaced with AAAAAA. Arrows indicate protein–DNA complex. Error bars indicate standard deviation ($n=5$ in (h), $n=3$ in (e) and (g)). Student's two tailed *t*-test was performed and statistically significant differences were indicated by *, $P<0.05$ or ***, $P<0.001$.

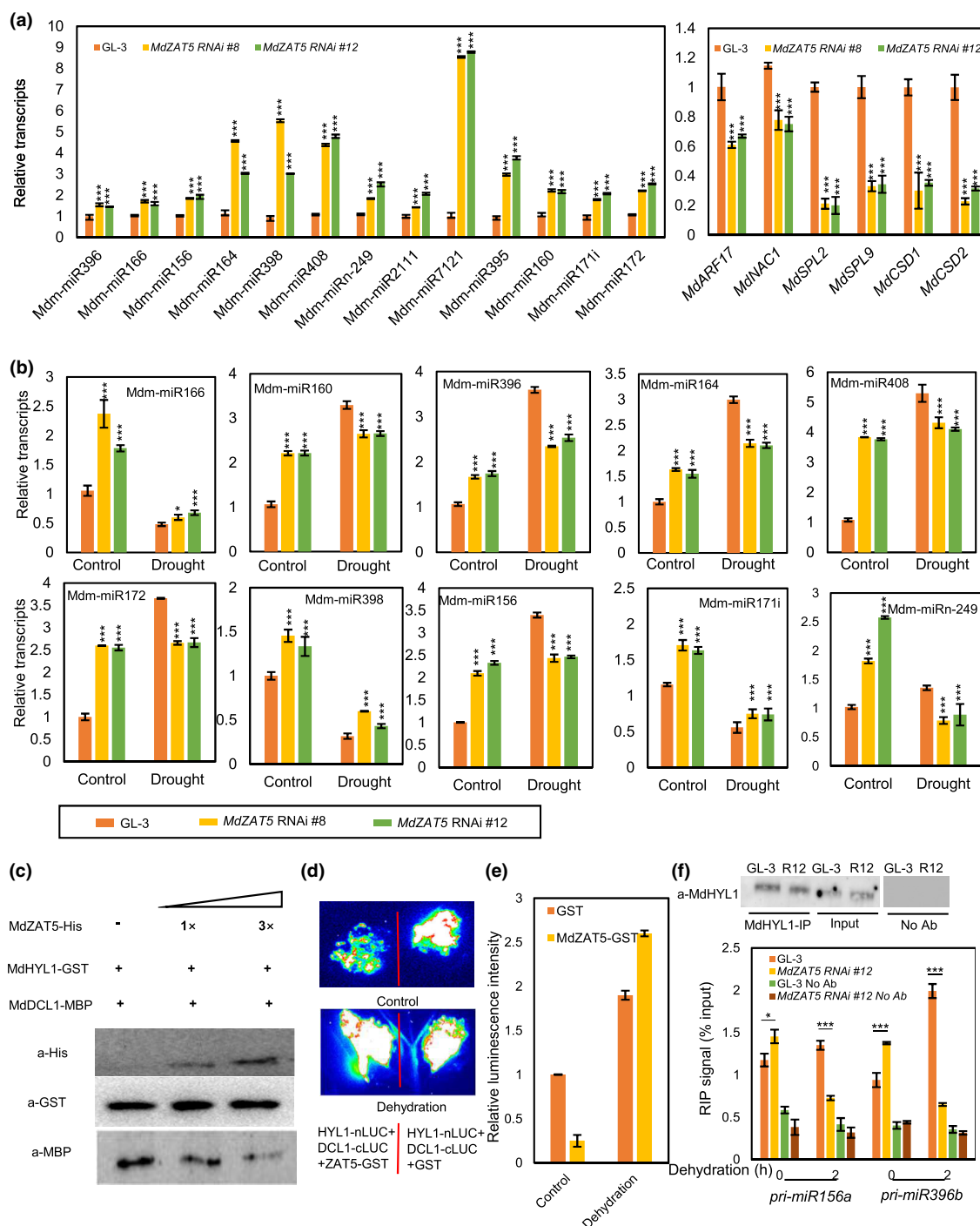


Fig. 8 MdZAT5 affects the abundance of microRNAs (miRNAs) under control and drought conditions. (a) The abundance of a group of miRNAs and their targets in apple (*Malus domestica*). (b) The abundance of Mdm-miR166, Mdm-miR160, Mdm-miR396, Mdm-miR164, Mdm-miR408, Mdm-miR398, Mdm-miR156, Mdm-miR171i, Mdm-miR172, and Mdm-miRn-249 in GL-3 and MdZAT5 RNAi transgenic plants under control and drought conditions for 7 d. (c) Competitive binding assays of MdZAT5 and MdDCL1 to MdHYL1. The mixture of MdZAT5-His and MdDCL1-MBP was added to immobilized MdHYL1-GST. The gradient indicates the increasing amount of MdZAT5-His. The experiments were repeated three times with similar results. A representative picture was shown here. (d) The effects of MdZAT5 on the interaction between MdHYL1 and MdDCL1 under control and drought by Luciferase (LUC) complementation imaging assays. MdHYL1-nLUC and MdDCL1-cLUC were co-expressed in tobacco leaves for 60 h, then glutathione S-transferase (GST) or MdZAT5-GST was injected and incubated for 30 min. (e) Analysis of the relative luminescence intensity shown in (d). Error bars denote standard deviations. (f) The association of MdHYL1 with Pri-miRNAs genes in GL-3 and MdZAT5 RNAi plants detected by RNA immunoprecipitation (RIP). Immunoprecipitation (IP) was performed with the anti-HYL1 antibodies. Detection of MdHYL1 after IP was shown in the upper panel. Furthermore 10% of IPs and 3% input proteins were detected by Western blot. The pri-miRNAs associated with MdHYL1 were examined by reverse transcription quantitative polymerase chain reaction (RT-qPCR) and normalized to the input. Error bars indicate standard deviation ($n = 3$). Student's two tailed t -test was performed and statistically significant differences were indicated by *, $P < 0.05$ or ***, $P < 0.001$.

We next were interested in whether the interaction of MdHYL1 with pri-miRNAs was affected by MdZAT5. The RNA immunoprecipitation (RIP) assay showed that abundance of MdHYL1-bound pri-miRNAs were enhanced in *MdZAT5* RNAi under normal conditions, while reduced in *MdZAT5* RNAi under drought stress, as compared to GL-3 (Fig. 8f). This result revealed that MdZAT5 may facilitate pri-miRNA loading to the DCL1 complex, thereby promoting miRNA processing under drought conditions, whereas it has an opposite effect under normal conditions.

MdHYL1 is essential for MdZAT5-mediated drought tolerance and miRNA biogenesis

To dissect the genetic interaction between *MdZAT5* and *MdHYL1*, we generated double transgenic apple calli with reduced expression of both *MdZAT5* and *MdHYL1* (*MdZAT5* RNAi/*MdHYL1* anti), as well as single transgenic calli with reduced *MdZAT5* (*MdZAT5* RNAi) or *MdHYL1* (*MdHYL1* anti) expression (Figs 9a, S23). The transgenic calli did not show any growth differences compared with control (the WT calli transformed with an empty vector) under control conditions (Fig. 9a,b). However, after PEG treatment for 15 d, *MdZAT5* RNAi and *MdHYL1* anti calli showed slower growth, and the relative growth rate of *MdHYL1* anti calli was lower than that of *MdZAT5* RNAi calli (Fig. 9a,b). Moreover, the double transgenic calli exhibited a similar growth rate to *MdHYL1* anti calli (Fig. 9b), indicating that MdHYL1 acts downstream of MdZAT5 in the drought stress response.

Under control conditions, knockdown of *MdHYL1* in *MdZAT5* RNAi calli rescued transcript levels of Mdm-miR160, Mdm-miR396, Mdm-miR164, Mdm-miR408, Mdm-miR156, Mdm-miR166, and Mdm-miR398, indicating that MdHYL1 is epistatic to MdZAT5 in miRNA biogenesis under control conditions. After PEG treatment, the abundance of Mdm-miR398, Mdm-miR166, and Mdm-miR160 in *MdZAT5* RNAi/*MdHYL1* anti was comparable to that in *MdHYL1*-anti calli, whereas the reduction in Mdm-miR164, Mdm-miR156, Mdm-miR396, and Mdm-miR408 transcripts in *MdZAT5* RNAi calli was more severe when *MdHYL1* expression was repressed (Fig. 9c). These results indicated that regulation of miRNA biogenesis by MdZAT5 is partially dependent on MdHYL1 under drought conditions.

MdZAT5 directly regulates the transcription of drought-responsive primary miRNAs

As shown in Fig. 8(a), all tested miRNAs were upregulated in *MdZAT5* RNAi plants. To further explore the molecular mechanism by which MdZAT5 influences miRNA biogenesis, we analyzed the expression levels of these 14 pri-miRNAs by RT-qPCR in *MdZAT5* RNAi and GL-3 plants. Some of the pri-miRNAs (pri-miR156, pri-miR395a, pri-miR164c, pri-miR166c, pri-miRn-249, pri-miR7121a, pri-miR172c, and pri-miR171i) were upregulated, whereas others (pri-miR398, pri-miR408a, pri-miR160e, pri-miR164e, pri-miR396b and pri-miR2111a) were

downregulated in *MdZAT5* RNAi plants (Fig. 10a). These results suggested that MdZAT5 acts as a transcriptional repressor in regulating the expression of pri-miR156, pri-miR395a, pri-miR164c, pri-miR166c, pri-miRn-249, pri-miR7121a, pri-miR172c, and pri-miR171i. Moreover, downregulation of pri-miR398, pri-miR408a, pri-miR160e, pri-miR164e, pri-miR396b and pri-miR2111a may result from the elevated expression of *MdHYL1* in *MdZAT5* RNAi plants (Fig. 7g), which led to faster processing of miRNA precursors.

Because MdZAT5 can bind to the promoters of its target genes, including *MdRHA2A*, *MdLEA14*, *MdCAT3*, and *MdTPX1*, we next wondered whether MdZAT5 could bind to MIR genes, especially those whose expression was upregulated in the *MdZAT5* RNAi plants (Fig. 10a). Promoter analysis showed that the promoters of *Mdm-miR171i* and *Mdm-miR172c* contained MdZAT5 recognition *cis*-elements (Figs S24, S25). We performed CHIP-PCR analysis and found that MdZAT5 could bind to the promoter regions of *Mdm-miR171i* and *Mdm-miR172c* (Fig. 10b–e). Electrophoretic mobility shift assays also showed that MdZAT5 could bind to the *Mdm-miR171i* and *Mdm-miR172c* promoter regions (Fig. 10f,g). Dual-LUC assays suggested that MdZAT5 could directly activate expression of *Mdm-miR172c* and repress *Mdm-miR171i* transcripts under dehydration conditions (Fig. S26), which was consistent with stem-loop qPCR results (Fig. 8b).

Mdm-miR171i plays a negative role in drought response in apple by regulating antioxidant enzyme genes (Wang *et al.*, 2020). To explore the effect of Mdm-miR172c in drought stress, we generated two lines of *Mdm-miR172c* OE transgenic calli (*Mdm-miR172c* OE #10 and OE #11) (Fig. S27). The transgenic calli did not show any growth differences compared with WT under control conditions (Fig. 10h,i). However, after PEG treatment for 15 d, *Mdm-miR172c* OE calli showed faster growth than WT, and the relative growth rate of *Mdm-miR172c* OE calli was higher than that of WT (Fig. 10h,i), indicating that Mdm-miR172c plays a positive role in the drought stress response.

Discussion

As a popular fruit crop, apple yield and quality are greatly damaged by extreme environmental conditions, among which drought is a major factor. Hence, the production of apple trees with improved drought tolerance is urgently needed, especially through biotechnology. In the current study, we identified MdZAT5 as a positive regulator of drought tolerance and a potential candidate gene for use in apple drought improvement. We found that MdZAT5 facilitates drought improvement by regulating drought-responsive messenger RNAs (mRNAs) involved in drought tolerance and ROS scavenging, as well as the biogenesis of drought-responsive miRNAs. Importantly, we showed that MdZAT5 regulates drought tolerance and miRNA abundance by interacting with MdHYL1 and directly binding to the promoter of *MdHYL1* (Fig. 11).

The role of C2H2-type ZFPs in response to abiotic stress has been widely reported (Wang *et al.*, 2019). Overexpression of *ZINC FINGER PROTEIN 252* (*ZFP252*) in rice increased plant

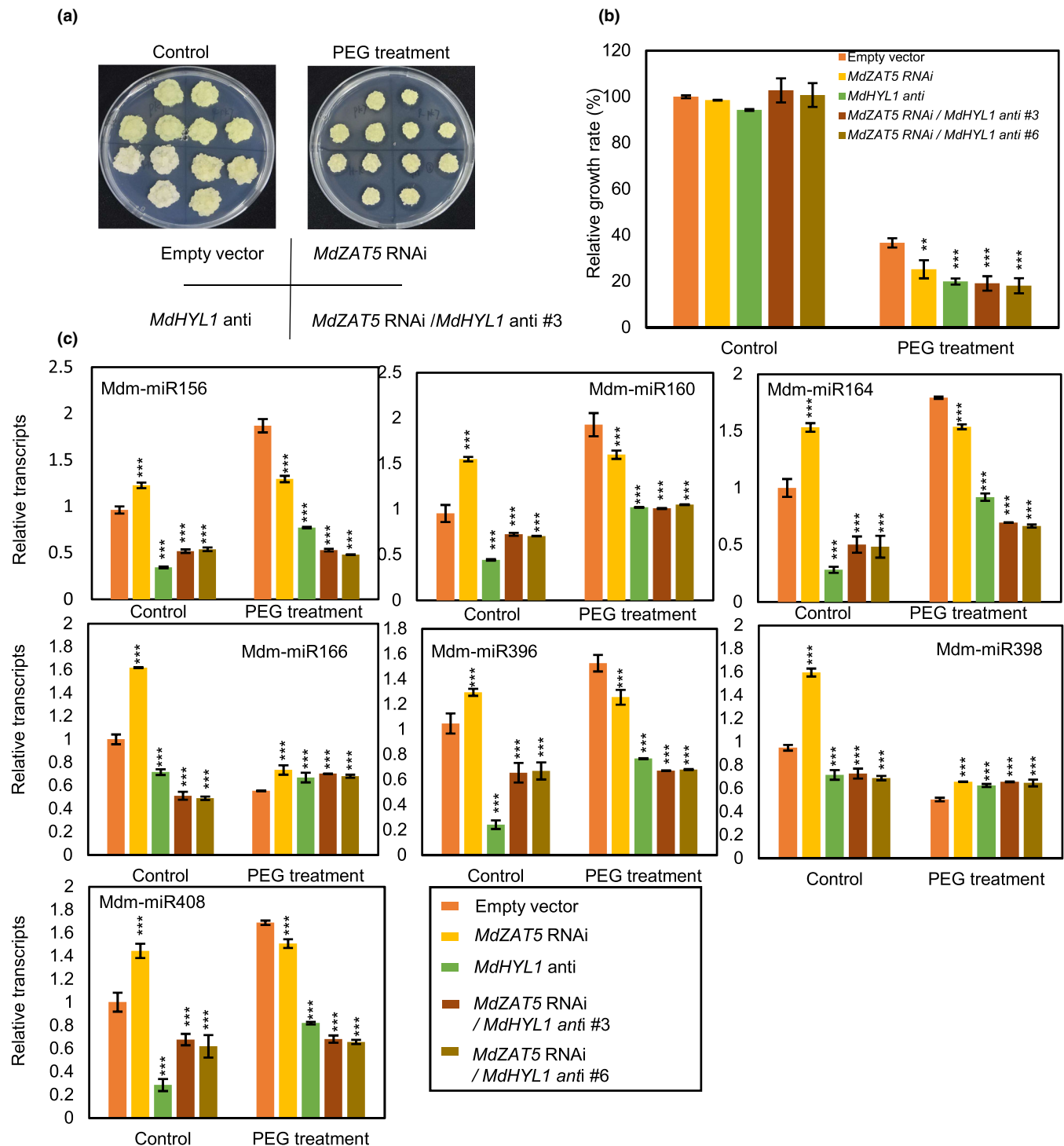


Fig. 9 MdHYL1 is essential for MdZAT5-mediated drought tolerance. (a, b) Morphological characteristics and relative growth rates of transgenic apple calli carrying the empty vector, *MdHYL1* anti, *MdZAT5* RNAi, or *MdZAT5* RNAi/*MdHYL1* anti in response to PEG treatment. Transgenic calli of equal weight were grown on Murashige & Skoog (MS) or MS media supplemented with PEG (−0.7 MPa) for 2 wk. (c) The abundance of microRNAs (miRNAs) in transgenic apple calli under control and PEG conditions. Empty vector, transformation of an empty vector into wild-type calli; *MdZAT5* RNAi/*MdHYL1* anti, knockdown of *MdHYL1* in *MdZAT5* RNA interference (RNAi) calli. Asterisks indicated values significantly different between the transgenic lines and the empty vector in each group (Control and PEG). Error bars indicate standard deviation ($n = 5$ in (b), $n = 3$ in (c)). Student's two tailed t -test was performed, and statistically significant differences are indicated by **, $P < 0.01$ or ***, $P < 0.001$.

survival rate under salt and drought stress by elevating the expression of stress-related genes (Xu *et al.*, 2008). *ZINC FINGER PROTEIN 245* (*ZFP245*) enhances drought tolerance in rice by

improving the plant's ability to scavenge ROS through SOD and POD (J. Huang *et al.*, 2009). However, the biological function of *ZAT5* in response to drought remains unclear, especially in

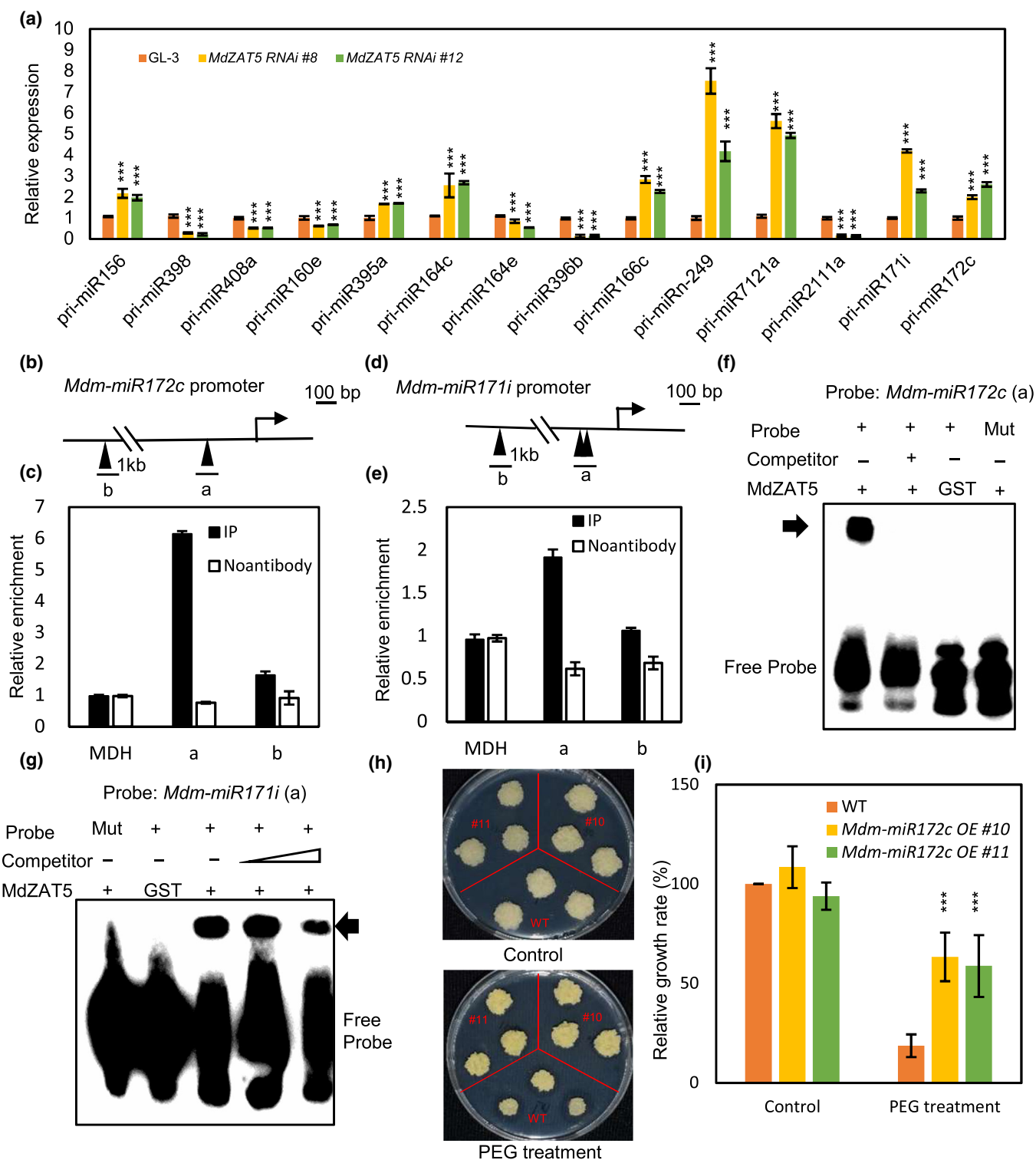


Fig. 10 *MdZAT5* directly regulates the transcription of several primary microRNAs (miRNAs). (a) The expression level of a group of primary miRNAs in GL-3 and *MdZAT5* RNA interference (RNAi) transgenic plants. (b) Diagram of *Mdm-miR172c* promoter. Fragment a contains one TCACTC motif; fragment b contains one TCACAA motif. (c) Chromatin immunoprecipitation quantitative polymerase chain reaction (ChIP-qPCR) analysis of *MdZAT5* bound to the promoter of *Mdm-miR172c*. MDH and no antibody serve as negative controls. (d) Diagram of *Mdm-miR171i* promoter. Fragment a contains one TCACTC, one ACACCT motif; fragment b contains one TCACTG motif. (e) ChIP-qPCR analysis of *MdZAT5* bound to the promoter of *Mdm-miR171i*. MDH and no antibody serve as negative controls. (f, g) Electrophoretic mobility shift assay (EMSA) analyses of *MdZAT5* protein bound to the promoter of *Mdm-miR172c* and *Mdm-miR171i*. (h) Morphological characteristics of wild-type (WT), *Mdm-miR172c* overexpressed (OE) transgenic apple calli after PEG treatment. (i) The relative growth rate of apple calli under PEG treatment. Wild-type and transgenic calli of equal weight were grown on Murashige & Skoog (MS) or MS media supplemented with PEG (−0.7 MPa) for 2 wk. Error bars indicate standard deviation ($n = 3$ in (a), (c), and (e), $n = 5$ in (i)). Student's two tailed t -test was performed, and statistically significant differences are indicated by ***, $P < 0.001$.

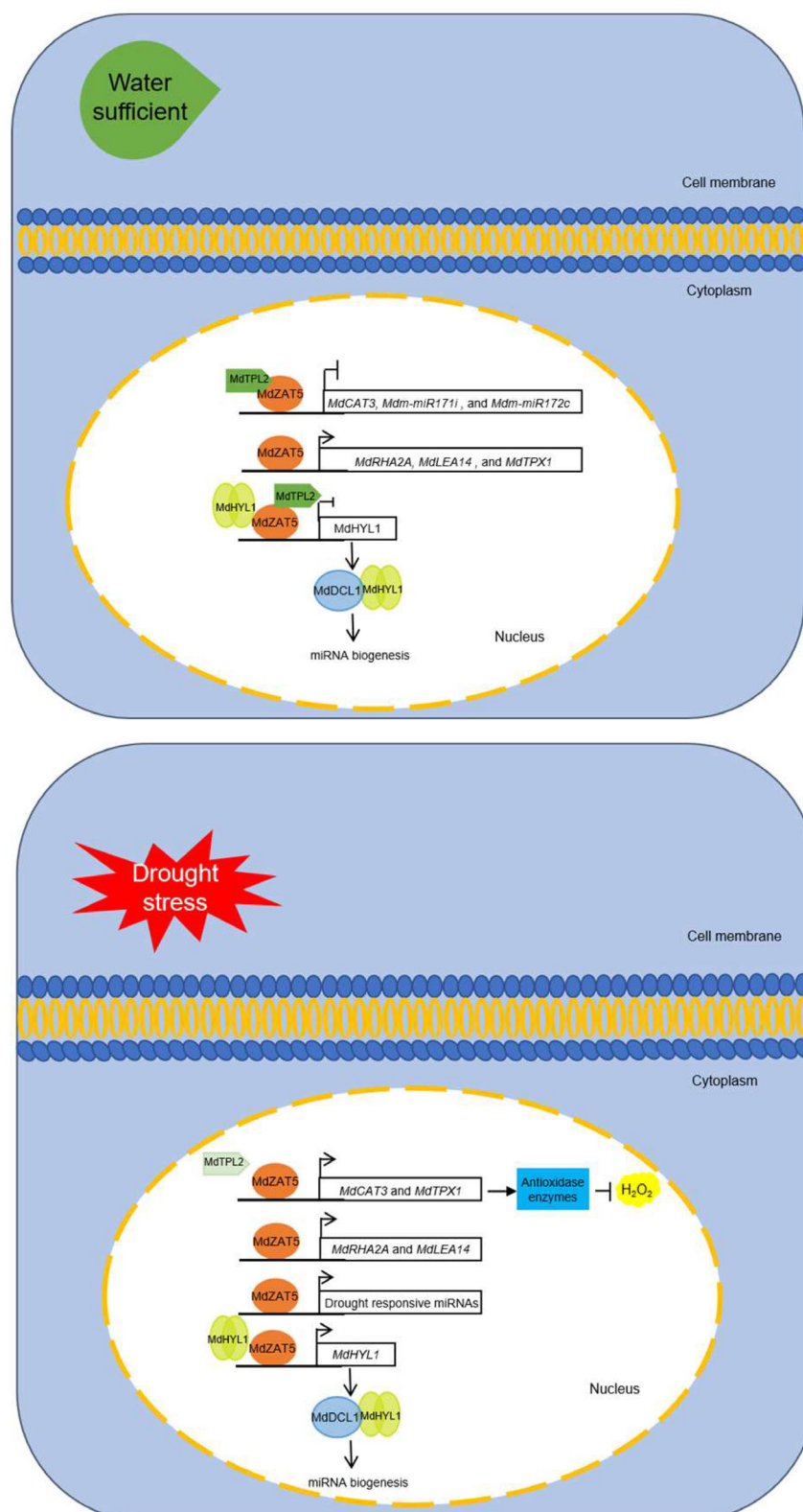


Fig. 11 A proposed model for MdZAT5 in response to drought in apple. Under water sufficient conditions, MdZAT5 represses the expression of genes including *MdCAT3*, *Mdm-miR171i*, *Mdm-miR172c*, and *MdHYL1*, probably through interacting with MdTPL2, as well as microRNA (miRNA) biogenesis by inhibiting the interaction between MdHYL1 and MdDCL1. In addition, MdZAT5 positively regulates its targets including *MdRHA2A*, *MdLEA14*, and *MdTPX1* by an unknown mechanism. Under drought stress, MdTPL2 is repressed, MdZAT5 is then released to activate the expression of its targets, including *MdRHA2A*, *MdLEA14*, *MdTPX1*, *MdCAT3*, *MdHYL1*, and drought-responsive miRNAs. The blunt ended arrow indicated inhibition, and the pointed arrow indicated activation.

apple trees under abiotic stress. In this study, we showed that the expression of *MdZAT5* was induced by short-term drought stress or PEG treatment (Fig. 1a,b). Moreover, *MdZAT5*-overexpressing plants showed reduced electrolyte leakage under drought treatment and enhanced tolerance to drought stress. By

contrast, *MdZAT5* RNAi plants were more sensitive to drought stress than GL-3 plants (Fig. 2). Drought stress induces an increase in ROS production (Gill & Tuteja, 2010). *MdZAT5* RNAi plants accumulated more H_2O_2 than GL-3 plants under short-term drought treatment because of their reduced ability to

scavenge ROS through CAT and POD. However, *MdZAT5*-overexpressing plants had lower H₂O₂ content and higher CAT and POD enzyme activity than GL-3 plants (Fig. 6). These findings demonstrated that *MdZAT5* plays an essential role in apple drought tolerance.

Zinc-finger proteins, including AZF1, AZF2, AZF3, and ZAT10/STZ, have been shown to bind specifically to A(G/C)T repeat sequences in their target promoters (Sakamoto *et al.*, 2004). We used DAP-Seq analysis and identified specific *cis*-elements for *MdZAT5* binding: TCACTC, TCACTA, TCACTG, ACACTC, ACACTA, ACACTG, TCACAC, TCACAG, TCACAA, and ACACAG, which were consistent with previous findings. Zinc-finger proteins have been reported to act as positive or negative regulators of their target genes. AZF1/2/3, ZAT7, ZAT10, and ZAT12 have been shown to function as transcriptional repressors because of the presence of an EAR motif (Sakamoto *et al.*, 2000, 2004; Ohta *et al.*, 2001; Hiratsu *et al.*, 2002; Ciftci-Yilmaz *et al.*, 2007; Kodaira *et al.*, 2011). Here, combining DAP-Seq and RNA-Seq data, we identified 391 genes that were both bound and regulated by *MdZAT5* (Fig. S28a). Among these genes, 126 (32%) were positively regulated by *MdZAT5*, whereas 265 (68%) were negatively regulated (Fig. S28b). These results indicated that *MdZAT5* acts as both an activator and a repressor of its direct target genes, although in most cases it functions as a negative regulator. Negative regulation by *MdZAT5* may be due to the EAR motif in its C terminus (Ohta *et al.*, 2001; Kazan, 2006; Han *et al.*, 2020). In addition, GO analysis of these 391 genes indicated that *MdZAT5* may function in responses to cold, salt, biotic stress, and hormones and participate in secondary metabolic pathways (Fig. S28c,d). Transcription factors containing the EAR motif can repress target genes expression via recruiting the co-repressors TPL/TPL-RELATED and histone deacetylases (Causier *et al.*, 2012; Wang *et al.*, 2013a; Ryu *et al.*, 2014). From our Y2H screening results, we identified the co-repressor MdTPL2 (Figs S17, S29). Y2H assay showed that *MdZAT5* interacted with MdTPL2 via the C-terminal EAR motif (Fig. S30a). Moreover, a split-LUC assay further confirmed MdTPL2 could interact with *MdZAT5* *in planta* (Fig. S30b). The RT-qPCR result showed that *MdTPL2* was repressed by drought stress (Fig. S30c). Hence, *MdZAT5* repressed the expression of its targets under normal conditions via recruiting MdTPL2. However, under drought conditions, *MdTPL2* was repressed, *MdZAT5* was released and to positively regulate its targets, though we cannot rule out the possibility that other factors may be involved in the positive regulation of *MdZAT5*. Furthermore, the ChIP-qPCR results showed that the binding sites on *MdHYL1* promoter of *MdZAT5* were different under normal and drought conditions (Fig. 7e), which may explain the negative and positive regulation of *MdHYL1* by *MdZAT5* under control and drought conditions, respectively.

The miRNAs play essential roles in plant drought response. In *Arabidopsis*, downregulation of miR165/166 can enhance drought tolerance (Yan *et al.*, 2016), and rice miR166 knock-down lines (STTM166) exhibit a rolled-leaf phenotype and enhanced drought tolerance (Zhang *et al.*, 2018). Moreover, miR172-overexpressing plants are more tolerant to drought than WT plants (Han *et al.*, 2013). In apple, knockout of Mdm-

miR171i in GL-3 improved drought tolerance by regulating antioxidant enzyme genes (Wang *et al.*, 2020). Furthermore, overexpression of Mdm-miR156, Mdm-miRn-249, Mdm-miR160e, or Mdm-miR172c increased the drought tolerance of apple (Niu *et al.*, 2019; Shen *et al.*, 2022; Fig. 10). Here, we found that *MdZAT5* positively regulated the abundance of miRNAs, including Mdm-miR160, Mdm-miR396, Mdm-miR164, Mdm-miR408, Mdm-miR156, Mdm-miRn-249, and Mdm-miR172, and negatively regulated Mdm-miR171i, Mdm-miR166, and Mdm-miR398 under drought stress (Figs 8, S21). This result indicated that *MdZAT5* may positively regulate drought tolerance through the regulation of these miRNAs.

The pri-miRNAs are processed by the dicing complex, which contains DCL1, HYL1, and SE (Dong *et al.*, 2008). HYL1 and SE are important for the accuracy of pri-miRNA processing. Moreover, the homodimerization of HYL1 is essential for the correct cleavage of pri-miRNAs (Yang *et al.*, 2014). In the *hyl1* mutant, pri-miRNAs are accumulated, and miRNA accumulation is decreased (Yang *et al.*, 2014). Here, we found *MdZAT5* may facilitate the interaction between MdHYL1 and MdDCL1 or pri-miRNAs, indicating the involvement of *MdZAT5* in miRNA processing. Indeed, we found that abundance of Mdm-miR398, Mdm-miR408, Mdm-miR164, Mdm-miR160, Mdm-miR396, and Mdm-miR2111 was higher in *MdZAT5* RNAi plants than GL-3, whereas their pri-miRNAs were reduced (Figs 8, 10, S20a, S21). In addition, the abundance of pre-miR408 in *MdZAT5* RNAi plants was less than that in GL-3 (Fig. S20), further indicating that *MdZAT5* plays a role in miRNA processing. Moreover, *MdZAT5* interacted with MdSE, MdDCL1, MdCDF2 *in vivo* (Fig. S22), further suggesting a role for *MdZAT5* in miRNA processing.

The promoters of most miRNA genes contain transcription factor binding motifs, indicating that the transcript levels of miRNAs are regulated by general and specific transcription factors (Xie *et al.*, 2005; Megraw *et al.*, 2006; Guan *et al.*, 2013; Yu *et al.*, 2017). NEGATIVE ON TATA LESS2 (NOT2) and the putative MYB domain-containing DNA-binding protein CELL DIVISION CYCLE 5 (CDC5) have been reported to promote transcription of miRNA genes (Wang *et al.*, 2013b; Zhang *et al.*, 2013). Moreover, cycling DOF transcription factor 2 (CDF2) binds directly to the promoters of some miRNAs and induces or represses these miRNA genes at the transcription level (Sun *et al.*, 2015). In this study, we determined that *MdZAT5* could regulate the transcription of a number of primary miRNAs, including pri-miR156, pri-miR395a, pri-miR164c, pri-miR166c, pri-miRn-249, pri-miR7121a, pri-miR172c, and pri-miR171i. The expression of these primary miRNAs was increased in *MdZAT5* RNAi plants and the abundance of these miRNAs were higher in *MdZAT5* RNAi plants than GL-3 (Figs 8, 10). In addition, the abundance of Mdm-miR172 and pre-miR172 in *MdZAT5* RNAi plants was more than that in GL-3 (Fig. S20). These results indicated that *MdZAT5* could negatively regulate transcription of miRNA genes. Furthermore, we found that *MdZAT5* binds directly to the promoters of *Mdm-miR171i* and *Mdm-miR172c* (Fig. 10), indicating that *MdZAT5* could regulate transcripts of pri-miRNAs by binding directly to their promoters.

In conclusion, we identified MdZAT5, a C2H2 ZFP that positively regulates apple drought tolerance. This positive regulation is achieved by mediating the expression of drought-responsive genes and miRNA biogenesis. The findings of the current study provide a useful strategy for increasing drought tolerance in apple and other fruit crops by altering the expression of one ZFP transcription factor gene using a transgenic approach.

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Author contributions

QG and CB designed the project. CB, GQ, FC, PC, CN, TR, XS, DZ, LM and FZ performed the experiments. JH and FM analyzed the data. CB and QG wrote the manuscript.

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Data availability

All the sequencing data in this study have been submitted to the NCBI database with the project ID PRJNA774137 (RNA-Seq and DAP-Seq).

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Protein alignment of ZAT5 from different plant species.

Fig. S2 Expression level of *MdZAT5* in *MdZAT5* transgenic plants (*Malus domestica*).

Fig. S3 PCA analysis of RNA-sequencing in apple (*Malus domestica*).

Fig. S4 The enriched Gene Ontology terms of differentially expressed genes in *MdZAT5* RNAi plants (*Malus domestica*) under simulated-drought stress.

Fig. S5 The correlation analysis of RNA-sequencing data and reverse transcription quantitative polymerase chain reaction analysis in *Malus domestica*.

Fig. S6 Verification of RNA-sequencing data using GL-3 and *MdZAT5* OE plants (*Malus domestica*) in response to drought stress conditions.

Fig. S7 *Malus domestica* protein MdZAT5 expression and purification.

Fig. S8 The enriched Gene Ontology terms for putative direct targets of MdZAT5 revealed by the DAP-Seq analysis in *Malus domestica*.

Fig. S9 Protein alignment of MdRHA2A and AtRHA2A.

Fig. S10 Promoter analysis of *MdRHA2A* in *Malus domestica*.

Fig. S11 Protein alignment of MdLEA14 and AtLEA14.

Fig. S12 Promoter analysis of *MdLEA14* in *Malus domestica*.

Fig. S13 Protein alignment of MdTPX1 and AtTPX1.

Fig. S14 Promoter analysis of *MdTPX1* in *Malus domestica*.

Fig. S15 Protein alignment of MdCAT3 and AtCAT3.

Fig. S16 Promoter analysis of *MdCAT3* in *Malus domestica*.

Fig. S17 Yeast two-hybrid library screening results of MdZAT5 in *Malus domestica*.

Fig. S18 Protein alignment of MdHYL1 and AtHYL1.

Fig. S19 Promoter analysis of *MdHYL1* in *Malus domestica*.

Fig. S20 The abundance of miRNAs detected by Northern blot in *Malus domestica*.

Fig. S21 The abundance of Mdm-miR160 detected by Northern blot in *Malus domestica*.

Fig. S22 *Malus domestica* protein MdZAT5 interacts with MdDCL1, MdCDF2 and MdSE but not with MdCDC5, MdAGO1, or MdNOT2 *in vivo*.

Fig. S23 Identification of *MdZAT5* RNAi, *MdHYL1* anti, or *MdZAT5* RNAi/*MdHYL1* anti transgenic apple calli.

Fig. S24 Promoter analysis of *Mdm-miR172c* in *Malus domestica*.

Fig. S25 Promoter analysis of *Mdm-miR171i* in *Malus domestica*.

Fig. S26 *Malus domestica* protein MdZAT5 can bind to *Mdm-miR171i* and *Mdm-miR172c* promoters by dual-luciferase assay.

Fig. S27 Identification of *Mdm-miR172c* OE transgenic apple calli.

Fig. S28 The direct target genes of MdZAT5 in *Malus domestica*.

Fig. S29 Protein alignment of MdTPL2 and AtTPL2.

Fig. S30 MdZAT5 interacts with MdTPL2 via the C-terminal EAR motif.

Methods S1 Supplementary materials and methods.

Table S1 Primers and vector used in this study.

Table S2 Information about RNA-sequencing libraries.

Table S3 Differentially expressed genes in drought-treated GL-3 compared with GL-3 under control conditions.

Table S4 Differentially expressed genes in control-treated *MdZAT5* RNAi plants compared with GL-3 plants under control conditions.

Table S5 Differentially expressed genes in drought-treated *MdZAT5* RNAi plants compared with GL-3 plants under drought conditions.

Table S6 The putative targets of MdZAT5 identified by DAP-Seq analysis.

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